Find the Missing Skill, Build the Data: Targeted Data Sets for $\pi_{0.5}$ on LIBERO-Plus

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***Abstract*—A strong robot policy fails on only a few factors of the scene. We find the skills that $\pi_{0.5}$ does not have on LIBERO-Plus, and we build one data set for each of two of them. We measure each step against base $\pi_{0.5}$ on one evaluator, and each change carries its interval from the 40 tasks. Our recovery clips from changed robot starts add $+16.90$ $[13.55, 20.26]$ points on robot start and change camera by -4.13 $[-7.77, -1.05]$ points. With identical prompts on both sides, an adapter stage in the vision encoder with our rendered camera views adds $+6.04$ $[2.72, 9.68]$ points on camera.**

We also find a defect in the released benchmark code: it adds file-name text to the instructions of 8,493 cases. Removing that text alone changes base $\pi_{0.5}$ by $+1.07$ $[0.14, 2.08]$ points on the seven-axis mean. Against base $\pi_{0.5}$ behind the same rewriter, our training adds $+5.19$ $[3.77, 6.75]$ points on identical prompts. As a test-time system behind a vocabulary-guided rewriter, our leaderboard entry scores 93.42 % $[93.03, 93.82]$, the mean of two passes over the 10,030 cases.

We also train OpenVLA on our data mix. With the rewriter on both sides, it changes robot start by $+60.00$ $[54.85, 64.78]$ points and the seven-axis mean by $+6.12$ $[2.09, 9.82]$ points. It loses on light: -9.81 $[-19.19, -1.90]$ points.

I. Introduction

LIBERO-Plus [9] tests a robot policy on 10,030 cases, and each case changes one factor: the camera, the robot start position, the language, the light, the background, the sensor noise or the object layout. A policy that scores near 95 % [9] on LIBERO [23] can lose many points when one factor changes [9]. The public $\pi_{0.5}$ LIBERO checkpoint, for example, succeeds on only 72.73 % of the camera cases (Section II).

We start from the public $\pi_{0.5}$ [28] policy fine-tuned on LIBERO, the pi05_libero checkpoint of openpi [27], which we call base $\pi_{0.5}$. Its score on each factor shows the skills that it does not have (Section II). We target two of them with data: recovery from a changed robot start, and camera perturbations. For the third, instruction handling, we add no training and rewrite the instruction at test time. Every step is a comparison with base $\pi_{0.5}$ on one evaluator.

Unless we say otherwise, a score is the LIBERO-Plus seven-axis mean: the mean of the seven factor scores, which is also the leaderboard's own score. A case-weighted score, the successes over all 10,030 cases, carries the label (total).

Three of our results should carry over to other policies and other benchmarks. The first is that targeted data works where the policy is weak. Our recovery clips add $+16.90$ points on robot start, on top of the benchmark's own training set, and they change camera by -4.13 points. That training set alone gives $+10.82$ points on camera. An encoder adapter stage with our rendered camera views then adds $+6.04$ points on camera (Table V gives both with their intervals).

The second is a defect in the released benchmark code, which appends file-name text to the instruction of 8,493 cases. Removing that text alone gains base $\pi_{0.5}$ $+1.07$ points (Table XVIII). The defect reaches any evaluation that sends the instruction as the benchmark builds it, but we measured its cost on $\pi_{0.5}$ models only. The third is the gain of our training itself, apart from the prompts: against base $\pi_{0.5}$ behind the same rewriter, our training adds $+5.19$ $[3.77, 6.75]$ points on identical prompts.

The leaderboard number comes second, and we label it so. The system that we submit, which we call the leaderboard system, puts a vocabulary-guided rewriter in front of our policy. This is a language model that rewrites each instruction into the vocabulary of the policy's 40 training instructions, without the instructions themselves. So the system is both a fine-tuned and a test-time system. It scores 93.42 % $[93.03, 93.82]$ over two passes, and Figure 7 traces its gain over base $\pi_{0.5}$ step by step. Section XI places that score among the reported ones. The order there is not a controlled comparison, and we chose this system as the headline after its first pass.

All our models are in the fine-tuned category of LIBERO-Plus. They train on its released training set, and our data targets its factors, so they fit the benchmark to a degree. Our closed-set language stage fits it most, because its only outputs are the 40 task instructions, and we report it as a side result (Appendix F).

Our contributions are:

1) Two data sets that we built, one for each of two missing skills: recovery clips from changed robot starts, and rendered camera views. We use them with the released training set of the benchmark, and we give the choices that we tested for each. On the closed-set line (from mxri-lg-1), an adapter stage in the vision encoder gains $+4.38$ $[2.11, 6.79]$ points on camera, and our views add $+3.38$ $[1.84, 5.00]$ points on top of it.
2) The file-name defect of the benchmark prompts, and its cost on $\pi_{0.5}$ models (Section VIII and Appendix F).
3) The gain of our training on identical prompts, apart from prompt handling. On the prompts that the closed-

set model acts on, it gains $+5.16$ points over base $\pi_{0.5}$ (Table XVIII).

4) A diagnosis of base $\pi_{0.5}$ on seven factors, which shows the factors where it is weak.
5) A vocabulary-guided rewriter for language at test time. Its rulebook holds the word list of the 40 training instructions, but not the instructions. Only 49.3 % of its rewrites of the language cases are one of the 40 instructions, and we detect no difference between it and a closed-set stage that we trained for the 40 tasks (Section VIII).
6) An accounting of the gain of the leaderboard system in four steps: the benchmark's training set, our recovery clips, the vocabulary-guided rewriter and the camera stage (Table V and Figure 7).
7) A fair setup (Section V). No rendered view is within 10 degrees of a test camera of its task, and the language data holds no test instruction. The closed-set stage gains on language also on prompts far from its training prompts. The rulebook of the vocabulary-guided rewriter holds no test instruction and none of the 40 task instructions.
8) Our full data mix on FLUX-3 and on OpenVLA (Section XII). The OpenVLA pair also differs in its start and its steps.

II. Base $\pi_{0.5}$ on LIBERO-Plus

We score base $\pi_{0.5}$, the public LIBERO checkpoint pi05_libero 7ec533af, with the protocol of Section IV. We converted it to our serving format and checked that all 812 tensors are bitwise equal to the source. It succeeds on 85.71 % on the seven-axis mean, and 84.72 % (total). Figure 1 compares these scores with those of the leaderboard system (Section VIII).

Three papers print one LIBERO-Plus row for $\pi_{0.5}$: 85.7 % (total) [14, 24, 33]. RoboHarness [13] gives the same total from the pi05_libero checkpoint, with different factor scores. Against that row, our pass of base $\pi_{0.5}$ is lower on robot start (-5.3 points) and camera (-3.1 points), and higher on language ($+2.8$ points). Those three papers do not name their checkpoint or evaluation settings, so we do not know if it is the one that we score.

The scores show where base $\pi_{0.5}$ is weak. It is weakest on the camera (72.73 %) and on the robot start (74.13 %). Language (86.14 %), object layout (86.89 %) and sensor noise (87.32 %) come next. Light (96.58 %) and background (96.19 %) are the strongest factors. We target the camera, the robot start and the language. Section XIII gives our results on layout.

The benchmark gives each case a level from 1 to 5 (Table I). On camera, robot start, language and sensor noise, base $\pi_{0.5}$ loses most at level 5. Camera goes from 97.0 % at level 1 to 44.4 % at level 5, and robot start from 98.4 % to 44.4 %. The gains of the leaderboard system are largest at the high levels. On camera, its gain is 0.5 points at level 1 and 36.0 points at level 5.

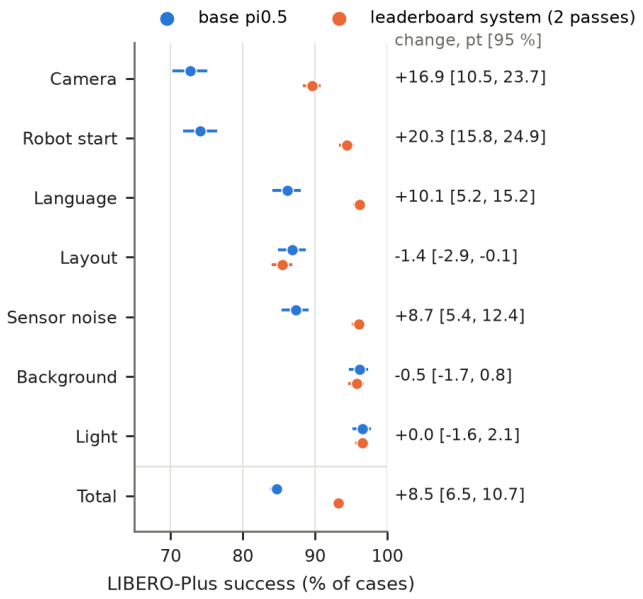


Fig. 1. Base $\pi_{0.5}$ and the leaderboard system on each test type of LIBERO-Plus and in total. The leaderboard system: the mean of its two passes. Dots: success, % of the cases of that type, with Wilson 95 % intervals over the episodes. Right: the change against base $\pi_{0.5}$, points, paired by case, with its 95 % interval from the 40 tasks. Total: all 10,030 cases. The seven-axis means are 85.71 and 93.42 %.

III. Related work and what we took from it

A. Benchmarks with changed scenes

LIBERO-Plus [9] changes one factor in each case, and LIBERO-PRO [42] changes objects, start positions, instructions and environments. *What we took:* we score each factor separately, because one total hides the missing skills.

B. Data from failures

LLM2LLM [20] makes new training data from the examples that a language model gets wrong. Skill-it [5] reweights the training mix toward the skills that a model does not know yet, and toward their prerequisites. MimicGen [25] makes new demonstrations from a small set of human ones. DART [19] adds noise to the demonstrator's actions, so that the demonstrations show how to recover from small errors. *What we took:* one data set for each missing skill, made in the simulator.

C. Camera changes

RoVi-Aug [4], VISTA [31] and RoboSplat [39] make new views of demonstrations with generative models or 3D reconstruction. AnyCamVLA [10] changes the test view back to the training view at test time. Other work changes the action space or adds 3D cues [1, 22].

Huang et al. [12] render LIBERO demonstrations again from the same simulator state with moved cameras. They fine-tune the policy with an added cross-view loss and mask the wrist camera. Feature Linear Adaptation [21] trains an adapter in the vision encoder of $\pi_{0.5}$ from one demonstration of a new view. *What we took:* same-state rendering from the first, and

TABLE I
SUCCESS BY FACTOR AND LEVEL

Factor	Level 1	Level 2	Level 3	Level 4	Level 5
Camera	97.0	92.2	79.7	71.5	44.4
	+0.5	+4.4	+11.5	+17.3	+36.0
Robot start	98.4	91.8	82.9	60.0	44.4
	−2.0	+3.8	+13.2	+33.2	+46.7
Language	93.0	95.0	96.5	91.8	59.1
	+4.7	+2.9	+1.2	+5.5	+32.2
Layout	94.3	92.7	91.8	82.1	70.6
	−2.0	−1.0	−2.4	+0.2	−1.6
Sensor noise	96.0	98.0	95.7	83.0	55.2
	+1.4	−0.4	+1.7	+14.1	+33.2
Background	98.3	98.5	97.3	95.7	87.0
	+0.2	−2.6	−0.8	+0.3	+2.1
Light (121 cases without a level)	95.5	97.5	97.5	96.7	93.8
	+2.3	+0.0	−0.8	−0.6	+1.8

Top line of each row: base $\pi_{0.5}$, % of the cases at that level. Bottom line: the change of the leaderboard system against it, points, paired by case, as the mean of its two passes. Levels: the benchmark's own labels. The 121 light cases without a level are not in a column. These readings are descriptive: we registered no rule per level. Each cell has 111 to 500 cases. The paired standard error of a change is 0.8 to 3.3 points, so a change of less than about two standard errors is not a reading of its own.

the encoder adapter from the second. We train the adapter on many rendered views with the action loss of the policy, and we keep the wrist camera.

D. Test-time systems

RoboHarness [13] routes subtasks between the public $\pi_{0.5}$ LIBERO checkpoint and a task-and-motion planner on LIBERO-Plus. A coding agent (Codex powered by GPT-5.5) decomposes tasks and assigns policies. Its setting permits "online adaptation and experience accumulation during evaluation". In its simulated experiments, the agent revises the orchestration code from execution traces, and the memory banks grow with successful executions. The paper does not state this for LIBERO-Plus separately, so that it applies there is our inference. *Our difference:* our system learns nothing during evaluation, because we fix its rewriter and its policy before the pass.

E. Language and prompts

DIAL [36] adds new instructions to robot demonstrations with vision-language models. Flex-π [38] reports that the released LIBERO-Plus code adds file-name text to the instructions. Rephrase Before You Act [34] rewrites each instruction once at deployment. A language model distils its rules from scored phrasings of a few training tasks. VLA Grounder [30] trains a rewriter with reinforcement learning for a frozen policy.

What we took: we report this defect and how our policy handles it (Section VIII). We also rewrite each instruction once before the policy acts, with rules that hold no task instruction of the benchmark.

IV. EVALUATION

We use LIBERO-Plus at commit `4976dc30`, MuJoCo 3.2.3, and one trial per case. A pass is one evaluation of all 10,030 cases. We report two metrics. The LIBERO-Plus case-weighted success rate, or total, is the successes divided by 10,030. The LIBERO-Plus seven-axis mean is the mean of the seven factor scores. A point is one percentage point of success.

Protocol. Each score of a $\pi_{0.5}$ model in this paper comes from one protocol (versions: Appendix D). It uses one fixed environment seed (42) and hosts with one GPU. The two passes of the start run (mxri-lg-1) also used some four-GPU hosts (CLAIMS.md, item 8). The policy uses 5 flow steps and 5 actions per query. It sends its raw gripper command, but the simulator uses only its sign, so a snapped command acts the same (CLAIMS.md, item 51). Before it acts, a model with our closed-set language stage maps the instruction to a task instruction with its own language model (Appendix F).

Intervals. We compare two models case by case on the same 10,030 cases, and a 95 % interval comes from that paired comparison. Where the text says "intervals from the 40 tasks", we resample the 40 base tasks that the cases come from, which gives wider intervals. A score of one system carries its interval over the cases. A change between two systems carries its interval from the 40 tasks.

Models. Table II names each model by its training stages, in training order, and a final number tells two runs of one recipe apart. In the text, seven models carry plain names:

- *base* $\pi_{0.5}$: pi05_libero, the public LIBERO checkpoint of openpi.
- *our fine-tune* (mxri-2): $\pi_{0.5}$ fine-tuned on the demonstrations and the recovery clips, with no language stage.
- *the closed-set recipe* (mxri-lg-2): our fine-tune, then the closed-set language stage. It is the reference of the ablations.
- *the start run* (mxri-lg-1): a run of the closed-set recipe that one of us released. The closed-set model starts from it.

- *the closed-set model* (mxri-lg-va-1): the camera stage on the start run. It was the headline of our first arXiv version (v1).
- *the leaderboard policy* (mxri-va): our fine-tune plus the camera stage, with no language stage.
- *the leaderboard system*: the leaderboard policy behind the vocabulary-guided rewriter of Section VIII.

We name the camera arms by what they change, as in Table IV. Most ablations of Sections IX and X ran on the closed-set model.

Noise. Two passes of mxri-lg-1 differ by 0.44 points on the seven-axis mean. Two seeds of the camera stage (seeds 2 and 3), both trained after a fix of our trainer's seeds, differ by $+0.19$ $[-0.31, 0.73]$ points. Each is one observed difference, not a noise floor for every comparison. The two runs of the closed-set recipe, the start run and mxri-lg-2, differ on the seven-axis mean. The start run has two passes, so the difference is 0.22 points with one pass and 0.65 points with the other. When the interval of a difference contains zero, we write that we detect no difference, which is no test of equivalence (Appendix A).

Test use. With our protocol, we scored 30 models on all 10,030 test cases in 31 passes, and base $\pi_{0.5}$ in two prompt controls (Appendix F). Most of these passes are the ablations and controls of this paper, or checks that it needed: seeds, a calibration, other policies. Two come from a side study that we do not report here. Our first arXiv version (v1) stands on these passes.

After v1, a layout screen scored 5 more models on the 1,525 layout cases only, and we do not report it. The 4 paraphrase arms of Appendix F then took one pass each on all cases. The vocabulary-guided rewriter added 10 passes: 3 on the 1,537 language cases and 7 on all cases. We stopped 5 more partway and read none of them. In all, 40 models ran on test cases under our protocol.

Of our 44 full passes, 15 informed the choice of a reported system and 10 the choice of a next experiment. Of the rest, 8 only measured, 4 measured noise, 6 scored other policies and 1 is not reported. Appendix H lists each pass with the decision that read it.

Other evaluation code scored at least 29 more passes on the test cases, for checks that we do not report (CLAIMS.md, items 21 and 56). In our first version, we chose the closed-set model as the headline for its rule against near-test views. We chose it after we scored the camera arms of Table IV. It is not the best of them on camera or on the seven-axis mean. We registered its comparison with the start run before training.

We made the leaderboard system the headline after its first pass, on the principle of Section XI. Five passes of our other models score above its mean (Section XI).

Outside these passes, test cases served three checks. A decoding gate on the 1,537 language test instructions came before the language stage. We registered its pass mark (95 %) in advance, and the stage reached 97.07 %. Later stages used held-out rewordings for their gates. A comparison on 840 libero_spatial test cases chose the start checkpoint: base $\pi_{0.5}$ scored 78 % and an earlier base 33 %. One robot-start test case checked the start mechanics of the recovery generator.

V. A FAIR SETUP

Three choices keep our gains away from the test set.

No near-test camera views. For the closed-set model (mxri-lg-va-1) and the leaderboard policy, we remove each rendered frame less than 10 degrees from a test camera of its task. This removes 46.2 % of the frames. The nearest kept view is 11.04 degrees from its test camera at the median. We did not measure the distance to test cameras of other tasks. The benchmark's broad data has camera angles 5 degrees from those of the test set [9], so the rule applies to our rendered views only.

The rulebook holds no test instruction. The rulebook of the vocabulary-guided rewriter holds rules and the word list of the 40 training instructions, but not the instructions themselves. Its author wrote it without reading the language test instructions, and we tuned its rules on 860 held-out paraphrases only. It also uses the kinds of perturbation in an earlier closed-set prompt of ours, and its rule 2 deletes file-name text by its form. At test time, the rewriter reads only the case's own instruction, which on six factors is a task instruction with the file-name text.

The rewriters themselves may have seen the public test instructions in pretraining (Section XIII). Appendix F gives the checks of the data of our closed-set stage.

One overlap remains: recovery start states. The recovery clips start from the 50 start states per task of LIBERO, each with an arm offset from our generator. The benchmark starts every case outside the layout factor from the first of those states. So 154 of the 6,441 logged successful episodes (2.4 %) start from the object placement of a test case, but with another arm offset. They give 0.53 % of the training frames of the start run, all from the return phase. We did not measure the effect of this overlap.

VI. SKILL: GENERAL PERTURBATIONS

The data. The benchmark authors released a training set that plays the LIBERO demonstrations again with changed camera, light, background, noise and objects (Figure 2). It has no changed robot starts, and it has only the 40 original instructions.

What we tested. We fine-tuned $\pi_{0.5}$ on the LIBERO-Plus training set only, with no clips and no language stage, and scored it against base $\pi_{0.5}$. Self-training gave no gain on a development set (CLAIMS.md, item 18).

Choice. Against base $\pi_{0.5}$, this data changes camera by $+10.82$ $[6.34, 15.46]$ points, light by -0.35 $[-3.34, 2.32]$, background by $+0.09$ $[-1.32, 1.55]$ and sensor noise by $+5.87$ $[3.04, 8.81]$ points. The broad data cannot teach the robot start or the language, because it does not contain them. Robot start changes by $+1.10$ $[-1.88, 4.07]$ points and language by -1.04 $[-2.91, 1.13]$ points. The interval of robot start contains zero, and that of language contains zero.

TABLE II
THE MODELS OF THIS PAPER

Name	Evaluator name	Start	What it adds	Use
base $\pi_{0.5}$	`pi05-libero`	–	the released LIBERO fine-tune [27]	every section
mx	`noclips-nolang`	base $\pi_{0.5}$	LIBERO-Plus demonstrations	General Perturbations
mx-lg	`noclips`	base $\pi_{0.5}$	mx, then the language stage	Robot Start
mxri-lg-1	`k0`	base $\pi_{0.5}$	demonstrations and recovery clips, then the language stage (released by one of us)	the start of the camera stage
mxri-2	`nolang`	base $\pi_{0.5}$	demonstrations and recovery clips (our run of the recipe)	Language
mxri-lg-2	`control`	base $\pi_{0.5}$	mxri-2, then the language stage	the reference of the ablations
mxri-va	`va-gpt`	mxri-2	the camera stage of mxri-lg-va-1, with no language stage	the leaderboard system, with the rewriter
mxri-lg-va-1	`fla-disjoint`	mxri-lg-1	an adapter stage on our views, near-test views removed	Camera; the closed-set model
mxri-lg-va-2	`fla-disjoint-s2`	mxri-lg-1	the same, another training seed	seeds
mxri-lg-va-3	`fla-disjoint-s3`	mxri-lg-1	the same, a third training seed	seeds
mxri-lg-ad	`fla-disjoint-nobank`	mxri-lg-1	the same adapter stage without the views	Camera
mxri-lg-vaf	`fla`	mxri-lg-1	an adapter stage on all our views	Camera
mxri-lg-vacv	`cvfla`	mxri-lg-1	the stage of mxri-lg-va-1 plus a cross-view loss	Camera
mxri-lg-vwcv	`cvsib`	mxri-lg-1	all weights but the language model, views and a cross-view loss	Camera
mxri-3	`ctlr1-s1`	base $\pi_{0.5}$	a third run of the demonstrations and recovery clips	Fair setup
mxri-lg-3	`ctlr1-lang`	base $\pi_{0.5}$	mxri-3, then the language stage on the cleaned file	Fair setup
mxrirw	`rew1`	base $\pi_{0.5}$	mxri-3's mix with reworded copies of the demonstrations	Language

Name: the model's training stages in training order. mx: LIBERO-Plus demonstrations. ri: recovery clips. lg: language stage. va: adapter stage with our rendered views, near-test views removed. ad: adapter stage without views. vw: views in a stage that trains all weights. cv: cross-view loss. rw: reworded copies of the demonstrations in the mix. f: the full set of views, none removed. A final number tells two runs of one recipe apart. Evaluator name: the name in our evaluation receipts. Section IV gives the plain names of the text, and Appendix D the training settings.

VII. SKILL: ROBOT START

The data. We record clips from changed start positions. Step 1: the robot starts with its joints moved by 0.1 to 0.5 radians. A script then moves the gripper back to its usual start pose. Step 2: a policy does the task from there, and we keep only the successes. Appendix D gives the script, the policy and the selection.

What we tested. We removed the recovery clips from the closed-set recipe and kept the language stage. Without the clips, robot start changes by -16.52 $[-19.76, -13.36]$ points. Before the language stage, the same removal changes it by -16.90 $[-20.26, -13.55]$ points. An earlier clip recipe of ours failed its development check (CLAIMS.md, item 17).

Choice. We use 6,480 clips, each the first 25 frames of a recovery. Each clip occurs 4 times, so the clips are 22.5 % of the training frames (clip frames). The scripted return acts in 7.4 % of the training frames (steer-back frames).

VIII. SKILL: LANGUAGE

The skill. Base $\pi_{0.5}$ succeeds on 86.14 % of the language cases. The benchmark rewords their instructions, and the policy trained on the 40 task instructions of LIBERO only. So the skill is to act on words that it did not train on.

Two routes. We tried two ways to handle the reworded instructions. The first is a test-time system: an external language model rewrites the text of each instruction before the policy acts. The second fine-tunes the language model of $\pi_{0.5}$ itself to map each instruction onto the 40 task instructions of LIBERO (Appendix F). This route measures how well $\pi_{0.5}$ learns to translate a perturbed instruction into one of its training instructions. We submit the first route to the leaderboard, because we expect a closed set of 40 instructions not to carry over to new tasks. A fine-tuning that gives $\pi_{0.5}$ its own rewrite for any instruction is a direction that we leave open (Section XIII).

A defect in the benchmark prompts. The released benchmark code adds file-name text to the instruction in the six factors other than language. This applies to 8,493 cases. LIBERO-Plus builds these instructions with the instruction-from-file-name rule of LIBERO (`benchmark/__init__.py`, lines 46 to 56, at commit 4976dc30). The LIBERO client of openpi [27] sends that string to the policy. Both routes remove that text.

The vocabulary-guided rewriter. A language model, GPT-6-astra, rewrites each benchmark instruction once, before the pass, and the policy acts on the rewrite. We call the rewriter vocabulary-guided because it rewrites each instruction into the vocabulary of the policy's training instructions. It reads the case's own instruction and a rulebook prompt, and no image. The rulebook holds rules and the word list of the 40 LIBERO training instructions: their objects, action templates and position words. It holds neither the 40 instructions nor a mapping onto them.

The vocabulary pulls the rewrites toward the 40 task instruc-

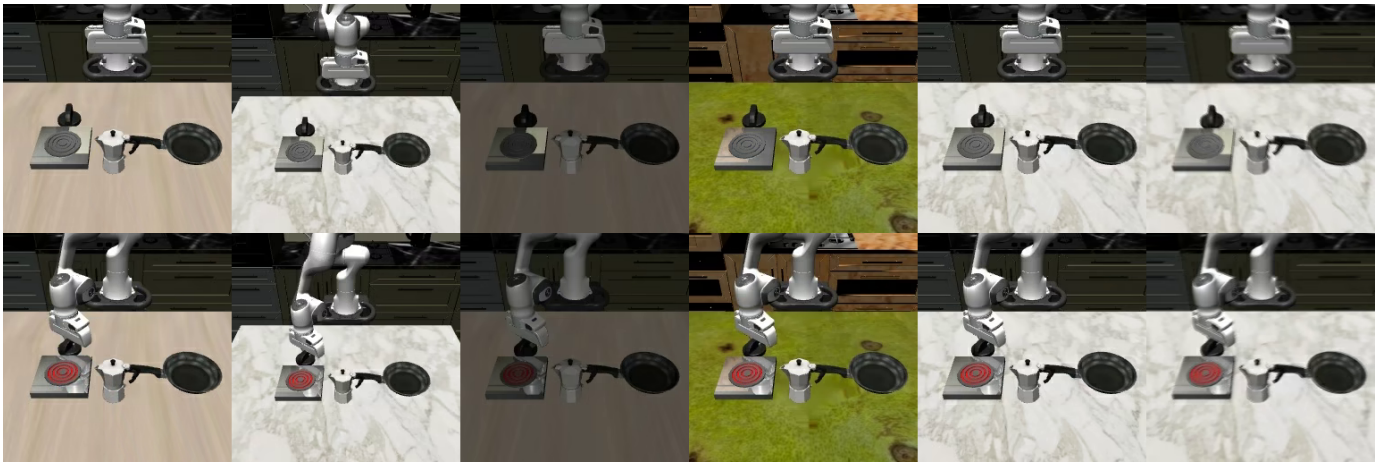


Fig. 2. Skill: General Perturbations. Left column: one LIBERO demonstration. Other columns: the same demonstration in the broad perturbation training set of LIBERO-Plus, with changed light, background and table. Top: the first frame. Bottom: a later frame.

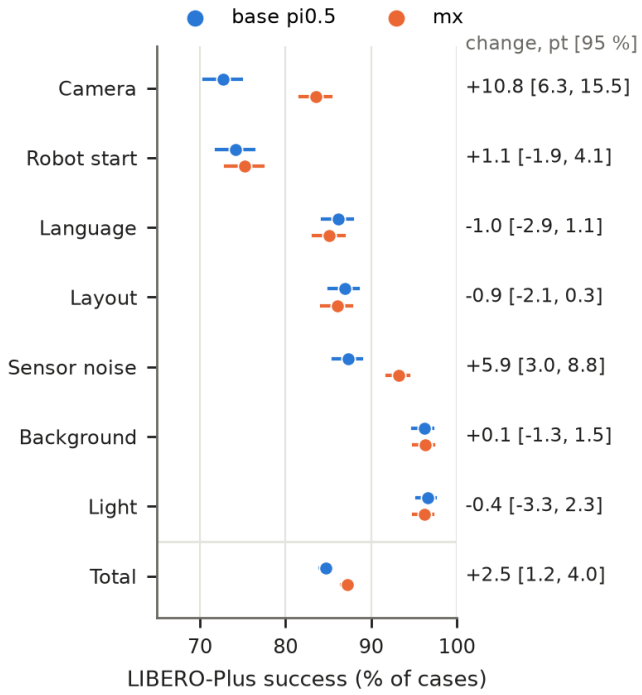


Fig. 3. Skill: General Perturbations. Base $\pi_{0.5}$ and mx, $\pi_{0.5}$ fine-tuned on the LIBERO-Plus training set only, on every test type. Success, % of the cases of that type, Wilson 95 % intervals. Right: the change paired by case, points, with its 95 % interval from the 40 tasks.

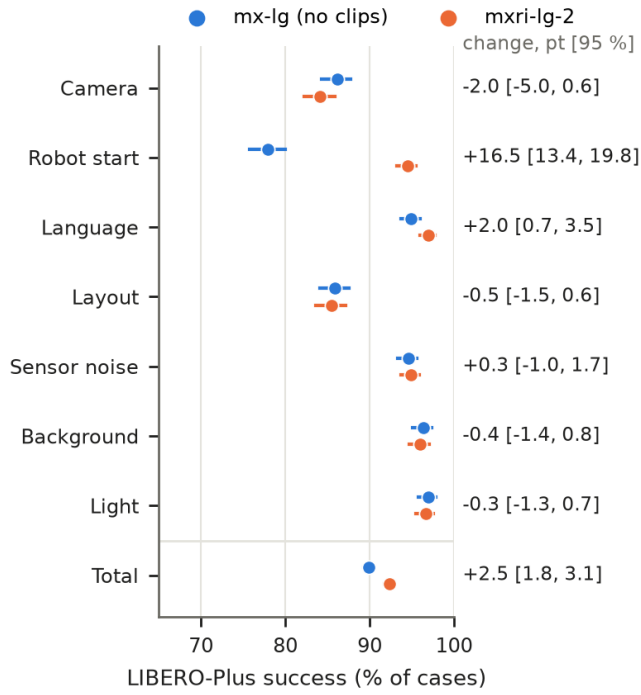


Fig. 4. Skill: Robot Start. The closed-set recipe (mxri-lg-2) and mx-lg, the same recipe without the recovery clips. Success, % of the cases of that type, Wilson 95 % intervals. Right: the change paired by case, points, with its 95 % interval from the 40 tasks.

tions: the rewriter knows their words, not their sentences. On the language cases, its output is mostly new text. Only 758 of the 1,537 GPT rewrites (49.3 %) are exactly one of the 40 task instructions, and 564 (36.7 %) for the open-weight Qwen rewriter below. The policy still acts on the rest. On the other six factors, the case's instruction is already a task instruction plus the file-name text. There, 8,473 of the 8,493 rewrites (99.76 %) equal the case's own task instruction, which shows the removal of the defect, not a closed set.

Section V gives how we kept the test instructions out of the rulebook. A language model that rewrites the prompt at test time is a test-time system. RoboHarness and TGL are test-time systems too (Table XI).

The leaderboard system. The leaderboard system is the rewriter in front of the leaderboard policy: our fine-tune (mxri-2) plus the camera stage of Section IX, with no language

stage. Its two passes, with no registered bar, score 93.51 % and 93.33 %. Their mean is 93.42 % $[93.03, 93.82]$, with its interval over the 10,030 cases. Over other sets of 40 tasks like these, the score could move more: its interval from the 40 tasks is 93.42 % $[91.09, 95.38]$. Against base $\pi_{0.5}$, the gain is $+7.71$ $[5.82, 9.75]$ points: camera $+16.85$ $[10.52, 23.67]$, robot start $+20.26$ $[15.76, 24.92]$, language $+10.05$ $[5.17, 15.20]$ and sensor noise $+8.71$ $[5.43, 12.39]$ (Figure 1).

Against the closed-set model (mxri-lg-va-1), the two-pass mean changes the seven-axis mean by $+0.19$ $[-0.38, 0.98]$ points and camera by -1.09 $[-2.39, 0.12]$ points. Without the camera stage, the same comparison is our fine-tune with the rewriter against the closed-set recipe (mxri-lg-2): -0.47 $[-1.09, 0.12]$ points. So, with or without the camera stage, we detect no difference between the two routes on the seven-axis mean.

The camera stage under the rewriter. The leaderboard system and our fine-tune with the same rewrite differ only by the camera stage. Section IX gives that pair.

Where the rewrite's gain comes from. On our fine-tune, the rewrite takes language from 85.23 % to 95.45 %, a change of $+10.21$ $[5.57, 15.18]$ points over the 1,537 language cases (Table III). We registered that criterion before any score. On all 10,030 cases, the rewrite changes the seven-axis mean of our fine-tune (89.62 %) by $+2.56$ $[1.72, 3.45]$ points. Removing the file-name text alone gives $+1.13$ $[0.45, 1.80]$ points of it, as on base $\pi_{0.5}$, and the rewrite adds $+1.43$ $[0.73, 2.18]$ points beyond that. No other factor drops by more than the registered 1.0 point against the removal alone.

The rewriter on other models. On the language cases, the leaderboard policy with the rewrite reaches 96.62 %. Against the closed-set model (mxri-lg-va-1) with its mapping, the change is $+1.17$ $[-0.75, 3.29]$ points, and against the closed-set recipe (mxri-lg-2) it is -0.33 $[-2.15, 1.33]$ points. The closed-set model with its stage off, whose weights still carry that stage's training, reaches 95.12 % with the rewrite. Against the same model with its mapping on, the change is -0.33 $[-1.93, 1.36]$ points. Section XII gives the rewriter on OpenVLA.

An open-weight rewriter. We ran one full pass of the leaderboard policy with Qwen3.5-9B, an open-weight model of 9 billion parameters, and the same rulebook: it scores 92.91 %. Against the two GPT passes, language changes by -3.32 $[-5.99, -0.72]$ and -2.21 $[-4.67, 0.14]$ points. The seven-axis mean changes by -0.60 $[-1.07, -0.18]$ and -0.42 $[-1.01, 0.16]$ points. So Qwen is about 2 to 3 language points below GPT. On the 860 held-out paraphrases, Qwen keeps the task in 87.6 % of them and GPT in 99.5 %, as Claude Sonnet 5.5 judges them.

The closed-set route. The closed-set model scores 93.23 %, and we detect no difference from the leaderboard system. Appendix F gives its data, its checks and its tests.

IX. Skill: Camera

The data. LIBERO-Plus moves the scene camera by an orbit, a change of distance, or a turn and tilt. We wrote our own code that makes the same camera moves. For each frame of the LIBERO human demonstrations, we put the simulator back into its recorded state. Then we render the frame again from a new camera (Figure 5). The label is the action that the human recorded. The data set has 270,685 frames, and the moves go up to 1.3 times the largest test move.

Where the data enters. We train only a LoRA [11] adapter in the vision encoder, as in Feature Linear Adaptation [21]. It sits on the MLP layers of the SigLIP [40] encoder of $\pi_{0.5}$ [28], and one third of each batch comes from the new views. As served, each parameter outside the adapter is bitwise equal to the model before this stage. Appendix D gives its settings.

No near-test views. The closed-set model (mxri-lg-va-1) and the leaderboard policy train on no rendered view near a test camera of the same task (Section V).

What we tested. Against the closed-set recipe (mxri-lg-2), the four adapter arms of Table IV gain 6.6 to 8.8 points on camera. The five arms that train all weights gain 3.0 to 3.8 points. These arms differ in more than the trained weights, so this comparison is not a one-factor test.

The camera stage in the leaderboard system. One pair isolates the camera stage: the leaderboard policy and our fine-tune (mxri-2), both with the same rewriter. The leaderboard policy starts from the weights of mxri-2 and adds only this stage. It changes camera by $+6.04$ $[2.72, 9.68]$ points and the seven-axis mean by $+1.25$ $[0.50, 1.97]$ points (Table V). The mxri-2 side has one pass and the other side two. In the closed-set model, the same stage follows the closed-set stage, but here it follows no language stage.

The views alone. We trained the same adapter stage without the rendered views (the no-views arm, mxri-lg-ad). The recipe's own data takes the third of each batch that held the views, so the volume of data is the same. Against this arm, the closed-set model gains $+3.38$ $[1.84, 5.00]$ points on camera. On the seven-axis mean, the change is -0.22 $[-0.92, 0.42]$ points, an interval that contains zero. So the views add camera skill, and we do not detect a further gain on the seven-axis mean.

Against the start run, the no-views arm gains $+4.38$ $[2.11, 6.79]$ points on camera and $+1.02$ $[0.38, 1.64]$ points on the seven-axis mean. This arm also ran a later version of our trainer, with fixed seeds, so it also changes the seeding code. So, in this pair, the gain on the seven-axis mean comes with the adapter stage and a change of trainer. We detect no further gain from the views.

Seeds 2 and 3 of the camera stage ran the same trainer and recipe as the no-views arm. Against it, they gain $+2.75$ $[0.88, 4.74]$ and $+2.50$ $[0.89, 4.19]$ points on camera. They change the seven-axis mean by -0.22 $[-0.78, 0.33]$ and -0.03 $[-0.51, 0.44]$ points, with intervals from the 40 tasks. The runs differ in their save interval and in where their spot machines resumed. Also, the no-views arm draws about 1.5 times more frames of the demonstrations and clips.

A cross-view loss. A cross-view loss on the camera stage lowered the score. Appendix G gives the arms and the cause.

TABLE III
THE VOCABULARY-GUIDED REWRITER ON THE 1,537 LANGUAGE CASES

Policy	Prompt	Language	Against as served	Against the mapping
mxri-2	as served	85.23 %	–	–
mxri-2	GPT rewrite	95.45 %	$+10.21$ $[5.57, 15.18]$	-1.50 $[-3.65, 0.42]$
mxri-lg-2	its mapping	96.94 %	–	–
mxri-va	GPT rewrite	96.62 %	$+11.39$ $[6.52, 16.62]^a$	$+1.17$ $[-0.75, 3.29]$
mxri-va	Qwen rewrite	93.43 %	–	-2.02 $[-4.24, 0.20]$
mxri-lg-va-1	GPT rewrite, mapping off	95.12 %	–	-0.33 $[-1.93, 1.36]$
mxri-lg-va-1	its mapping	95.45 %	–	–

Language: success, % of the 1,537 language cases. Changes in points, paired by case, with intervals from the 40 tasks. Against the mapping: a policy with the closed-set stage and its mapping (mxri-lg-2 for mxri-2, and mxri-lg-va-1 for the other rows). [a]against mxri-2 as served, so the change also holds the camera stage. GPT and Qwen read the case's instruction and the rulebook only. The Qwen row comes from its full pass, the other rewrite rows from passes on the language cases. On language, the two full passes of the leaderboard system give 96.75 % and 95.64 %, and the full pass of mxri-2 with the rewrite gives 95.38 %. Figure 1 and Table V read those full passes. A dash: not applicable.

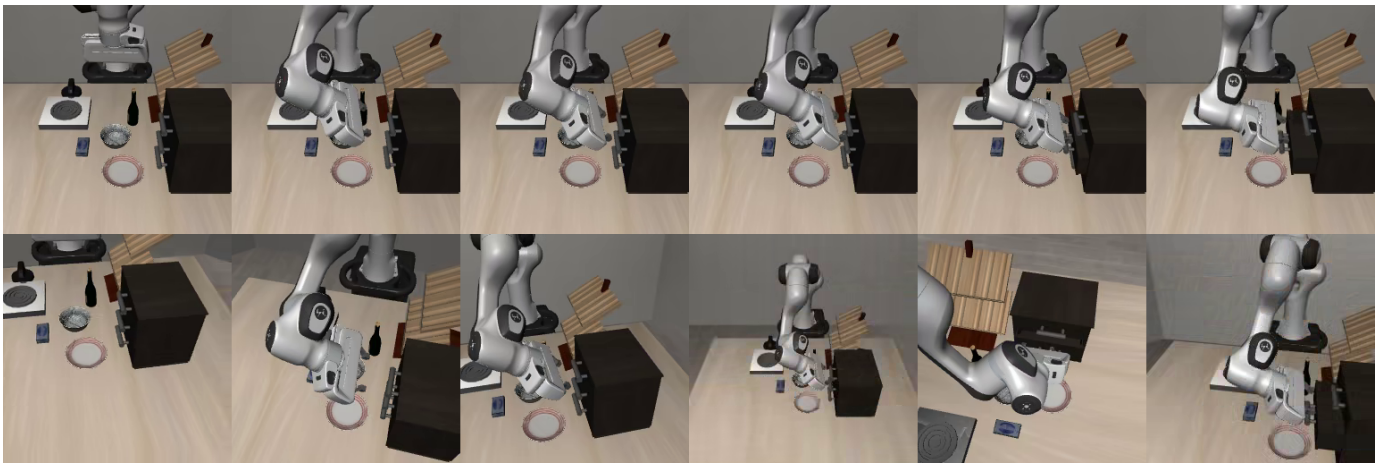


Fig. 5. Skill: Camera. One demonstration of our camera data. Top: the original camera at six moments of the demonstration. Bottom: our rendered views of the same simulator states, each from a moved camera.

Camera types. Against the start run, the camera gain of the closed-set model comes from the orbit moves: $+11.79$ points $[9.32, 14.27]$ on 992 cases. That interval is over cases, because we have none from the 40 tasks per move type. The distance and the turn and tilt moves show no clear gain (CLAIMS.md, item 4). The rule removed every rendered frame of the distance type.

Cost. Against the recipe before it, each adapter arm loses 0.7 to 1.5 points on language. Each arm that trains all weights loses 1.7 to 2.4 points.

X. THE FULL RECIPE AND ABLATIONS

Where the gain comes from. Table V and Figure 7 split the gain of the leaderboard system over base $\pi_{0.5}$ into four steps. The benchmark's training set gives $+2.23$ $[0.99, 3.61]$ points, mostly on camera and sensor noise. Our recovery clips add $+1.67$ $[0.86, 2.41]$, mostly on robot start, and they cost camera points. The vocabulary-guided rewriter adds $+2.56$ $[1.72, 3.45]$, mostly on language, and the camera stage $+1.25$ $[0.50, 1.97]$, mostly on camera. Together they give $+7.71$ $[5.82, 9.75]$ points.

Part of the rewriter's step comes from the file-name defect. On our fine-tune, removing that text alone gives $+1.13$ $[0.45, 1.80]$ points (Section VIII).

The ablations of this section ran on the closed-set line (mxri-lg-*), which shares its data and camera stages with the leaderboard system. Table VI gives the gain of the closed-set model (mxri-lg-va-1) against each of its starts, with intervals from the 40 tasks. Against base $\pi_{0.5}$, it gains $+7.52$ $[5.65, 9.52]$ points on the seven-axis mean and $+8.37$ $[6.40, 10.46]$ points (total). The camera stage alone gains $+7.75$ $[4.69, 11.07]$ points on camera against the start run, and $+6.57$ $[3.21, 10.29]$ against the closed-set recipe (mxri-lg-2), our rerun of it. The two readings share the closed-set model, so they are not independent. On the seven-axis mean, the camera stage gains $+0.80$ $[0.06, 1.53]$ and $+0.59$ $[-0.34, 1.48]$ points, and the second interval contains zero.

Table VII reads each step of the recipe against base $\pi_{0.5}$, and Table VIII removes one data set at a time. Most of the gain on sensor noise comes before the camera stage: the broad data alone changes sensor noise by $+5.87$ $[3.04, 8.81]$ points.

On light, background and layout, the leaderboard system changes by $+0.00$ $[-1.62, 2.07]$, -0.46 $[-1.72, 0.76]$ and

TABLE IV
SKILL: CAMERA. EACH CAMERA ARM AGAINST BASE $\pi_{0.5}$

Arm	What trains	Camera	Lang.	7-axis
mxri-lg-va-1 (no rendered view near a test camera)	encoder adapter	$+17.95$ [11.43, 24.97]	$+9.30$ [4.27, 14.63]	$+7.52$ [5.65, 9.52]
all rendered views	encoder adapter	$+20.14$ [13.36, 27.32]	$+9.37$ [4.11, 14.81]	$+8.09$ [6.12, 10.24]
adapter first, then full fine-tune	encoder adapter, then the rest	$+19.51$ [12.37, 27.07]	$+10.08$ [4.51, 15.82]	$+8.03$ [6.16, 10.09]
full fine-tune first, then adapter	all weights, then encoder adapter	$+19.20$ [12.30, 26.57]	$+9.37$ [4.23, 14.91]	$+8.32$ [6.25, 10.55]
rendered views at 22.5 % of frames	all weights	$+14.51$ [8.69, 20.44]	$+8.39$ [3.00, 13.97]	$+6.38$ [4.54, 8.34]
rollouts with labels from a teacher policy, 22.5 %	all weights	$+15.13$ [9.23, 21.27]	$+8.46$ [2.84, 14.29]	$+6.53$ [4.50, 8.67]
failed rollouts with labels from a teacher policy, 16.2 %	all weights	$+14.76$ [8.78, 21.12]	$+9.11$ [3.54, 14.94]	$+6.68$ [4.71, 8.86]
rendered views and new layouts, 22.5 % each	all weights	$+14.38$ [8.78, 20.39]	$+8.91$ [4.07, 14.12]	$+6.42$ [4.63, 8.36]
new layouts rendered from moved cameras	all weights	$+14.88$ [8.70, 21.51]	$+8.98$ [3.80, 14.42]	$+6.66$ [4.83, 8.64]

Change in points, paired by case, with 95 % intervals from the 40 tasks. One pass and one training run for each arm. The arms differ in more than one factor, and Table VIII gives the one-factor contrasts.

TABLE V
WHERE THE GAIN OF THE LEADERBOARD SYSTEM COMES FROM

Change	Source	Robot	Lang.	Camera	7-axis
the benchmark's training set, on the raw prompts	training	$+1.10$ [-1.88, 4.07]	-1.04 [-2.91, 1.13]	$+10.82$ [6.34, 15.46]	$+2.23$ [0.99, 3.61]
our recovery clips (our fine-tune, mxri-2)	training	$+16.90$ [13.55, 20.26]	$+0.13$ [-1.26, 1.50]	-4.13 [-7.77, -1.05]	$+1.67$ [0.86, 2.41]
the vocabulary-guided rewriter, on our fine-tune	rewriter	$+3.42$ [1.44, 5.44]	$+10.15$ [5.51, 15.14]	$+4.13$ [1.69, 6.92]	$+2.56$ [1.72, 3.45]
the camera stage, under the rewriter	training	-1.16 [-2.47, 0.07]	$+0.81$ [-0.28, 2.05]	$+6.04$ [2.72, 9.68]	$+1.25$ [0.50, 1.97]
all four: the leaderboard system against base $\pi_{0.5}$	all	$+20.26$ [15.76, 24.92]	$+10.05$ [5.17, 15.20]	$+16.85$ [10.52, 23.67]	$+7.71$ [5.82, 9.75]

Each row adds one step to the row above it: base $\pi_{0.5}$, then the benchmark's training set (on the raw prompts), then our recovery clips (our fine-tune, mxri-2), then the vocabulary-guided rewriter, then the camera stage (the leaderboard policy with the rewriter, the mean of two passes). The four rows add up to the last one, within rounding. Change in points, paired by case, with intervals from the 40 tasks.

TABLE VI
THE CLOSED-SET MODEL AGAINST EACH OF ITS STARTS, AND TWO MORE SEEDS

Model	Start		Camera	Seven-axis mean	Total
mxri-lg-va-1	base $\pi_{0.5}$	the base model	$+17.95$ [11.43, 24.97]	$+7.52$ [5.65, 9.52]	$+8.37$ [6.40, 10.46]
mxri-lg-va-1	mxri-lg-1	where it starts	$+7.75$ [4.69, 11.07]	$+0.80$ [0.06, 1.53]	$+0.97$ [0.20, 1.73]
mxri-lg-va-1	mxri-lg-2	our rerun of that recipe	$+6.57$ [3.21, 10.29]	$+0.59$ [-0.34, 1.48]	$+0.79$ [-0.17, 1.74]
mxri-lg-va-2	mxri-lg-1	where it starts	$+7.13$ [3.84, 10.82]	$+0.80$ [0.08, 1.54]	$+0.89$ [0.16, 1.65]
mxri-lg-va-3	mxri-lg-1	where it starts	$+6.88$ [3.91, 10.13]	$+0.99$ [0.27, 1.75]	$+1.11$ [0.39, 1.89]

Change of mxri-lg-va-1 against each start, points, with intervals from the 40 tasks. Camera: the camera factor. Seven-axis mean: the default score of this paper. Total: all 10,030 cases. We registered the comparison with mxri-lg-1 before training. Last two rows: two more training seeds of the camera stage. Seed 1, mxri-lg-va-1, trained before a fix of our trainer's seeds, and seeds 2 and 3 after it.

-1.44 $[-2.89, -0.10]$ points against base $\pi_{0.5}$ (Figure 1). The layout interval does not contain zero.

Two more seeds of the camera stage, mxri-lg-va-2 and mxri-lg-va-3, gain $+0.80$ and $+0.99$ points on the seven-axis mean against the start run (Table VI).

XI. REPORTED SCORES IN CONTEXT

This section places the leaderboard number among the scores that other work reports, as an order of reported scores, not a controlled comparison. The leaderboard system scores 93.42 % $[93.03, 93.82]$, the mean of two passes, with its interval over the cases. That is above every LIBERO-Plus score that other work reports, as of October 2026. The best entries of the public AllenAI leaderboard are RoboHarness and TGL at 92.4 % (Section III). The best other fine-tuned model is ZimaBlue at 92.0 %. Appendix B gives every reported score of at least 90 %.

Our margin over the best other score is 1.02 points, and 0.93 and 1.11 points for the two passes. The margin has no interval, because the other scores carry none and each comes from another evaluator. We measure our own score to about 0.4 points over the cases. Over other sets of 40 tasks like these, it could move by about 2 points (Section VIII).

We chose this system as the headline after its first pass, and our use of the test set is large (Section IV). Five passes of our other models score above its mean, up to 94.04 %. They are three camera arms (Table IV), the no-views arm (Section IX) and a control of a side study. With that much choice among our own passes, a margin of 1.02 points is within the range of selection effects. We chose the system for a principle, not for its score: it uses the words of the 40 task instructions, not the instructions. The order does not use the five passes above it.

The headline also depends on a closed model behind an

TABLE VII
EACH STEP OF THE CLOSED-SET RECIPE AGAINST BASE $\pi_{0.5}$

Step	7-axis (%)	Robot	Lang.	Camera	7-axis
base $\pi_{0.5}$	85.71	–	–	–	–
+ broad perturbation data	87.95	$+1.10$ [−1.88, 4.07]	-1.04 [−2.91, 1.13]	$+10.82$ [6.34, 15.46]	$+2.23$ [0.99, 3.61]
+ recovery clips	89.62	$+18.00$ [14.16, 21.83]	-0.91 [−3.08, 1.19]	$+6.69$ [2.27, 11.48]	$+3.91$ [2.61, 5.26]
+ language stage	92.64	$+20.32$ [15.88, 25.02]	$+10.80$ [5.78, 16.17]	$+11.38$ [6.41, 16.62]	$+6.93$ [5.35, 8.68]
+ encoder adapter stage with rendered views (ours)	93.23	$+20.84$ [16.39, 25.59]	$+9.30$ [4.27, 14.63]	$+17.95$ [11.43, 24.97]	$+7.52$ [5.65, 9.52]

Column 2: success rate, % of cases, seven-axis mean. Other columns: cumulative change against base $\pi_{0.5}$, points, paired by case, with intervals from the 40 tasks. We fixed each pairing before we scored base $\pi_{0.5}$. Last row: the encoder adapter stage with the rendered views. The views alone add $+3.38$ points on camera (Section IX). The last row starts from the start run (mxri-lg-1), a separate run of the recipe, not from the row above it (Table II).

TABLE VIII
REMOVE ONE DATA SET

Change	vs	Robot	Lang.	Camera	7-axis
remove all (base $\pi_{0.5}$)	mxri-lg-2	-20.32 [−25.02, −15.88]	-10.80 [−16.17, −5.78]	-11.38 [−16.62, −6.41]	-6.93 [−8.68, −5.35]
remove the recovery clips	mxri-lg-2	-16.52 [−19.76, −13.36]	-2.02 [−3.46, −0.74]	$+2.00$ [−0.61, 5.03]	-2.25 [−2.84, −1.69]
remove the language stage	mxri-lg-2	-2.32 [−4.45, −0.25]	-11.71 [−17.10, −6.63]	-4.69 [−6.97, −2.71]	-3.02 [−4.12, −1.99]
remove the clips, before the language stage	mxri-2	-16.90 [−20.26, −13.55]	-0.13 [−1.50, 1.26]	$+4.13$ [1.05, 7.77]	-1.67 [−2.41, −0.86]
remove the views, keep the adapter	mxri-lg-va-1	$+0.52$ [−0.72, 1.69]	-0.07 [−1.25, 1.04]	-3.38 [−5.00, −1.84]	$+0.22$ [−0.42, 0.92]
the camera stage (adapter and views)	mxri-lg-2	$+0.52$ [−0.83, 1.90]	-1.50 [−3.20, 0.00]	$+6.57$ [3.21, 10.29]	$+0.59$ [−0.34, 1.48]
cross view: the loss without the adapter	mxri-lg-va-1	-0.84 [−1.94, 0.32]	-0.91 [−2.53, 0.60]	-6.50 [−10.04, −3.27]	-2.13 [−3.07, −1.27]
cross view: the loss on the adapter stage	mxri-lg-va-1	-35.16 [−42.01, −28.68]	-18.15 [−25.72, −11.75]	-11.69 [−17.03, −6.85]	-17.57 [−22.94, −12.79]

Change of each model against the model in column vs, points, paired by case, 95 % intervals from the 40 tasks. Table II gives each name. We fixed each pairing before we scored these models.

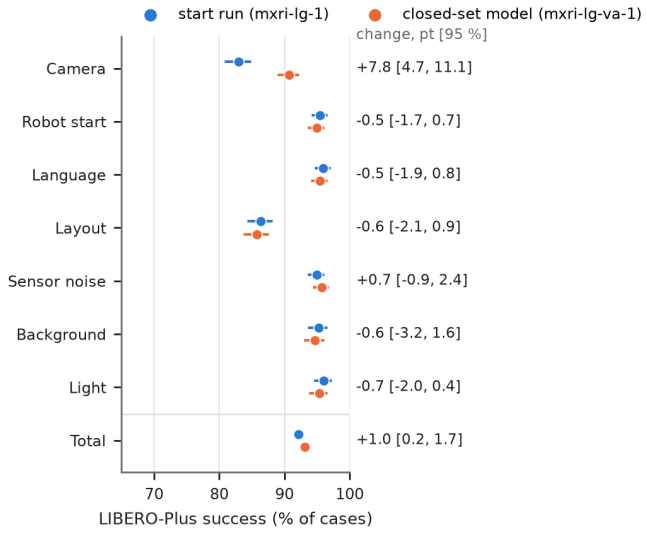


Fig. 6. Skill: Camera. The camera stage: the start run (mxri-lg-1) and the closed-set model (mxri-lg-va-1), which adds the encoder adapter stage with the rendered views. Success, % of the cases of that type, Wilson 95 % intervals. Right: the change paired by case, points, with its 95 % interval from the 40 tasks. The views alone give $+3.38$ of the camera gain (Section IX).

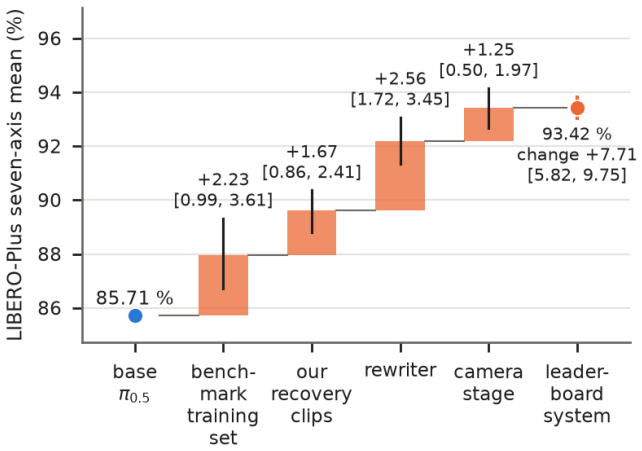


Fig. 7. Where the gain of the leaderboard system comes from: the seven-axis mean, from base $\pi_{0.5}$ through the four steps of Table V to the leaderboard system. Rewriter: the vocabulary-guided rewriter. Bars: the change of each step, points, paired by case, with its 95 % interval from the 40 tasks. Ends: the scores, % of the cases; the leaderboard system is the mean of its two passes, with its interval over the cases, and its total change carries its interval from the 40 tasks. The vertical axis does not start at zero.

API, GPT-6-astra. With the open-weight Qwen3.5-9B and the same rulebook, the leaderboard policy scores 92.91 % (Section VIII).

The leaderboard system trains on the benchmark's data and calls a language model at test time. So it is in the fine-tuned and test-time category. The protocol of the public AllenAI leaderboard is zero-shot from LIBERO, and the leaderboard system is outside it. RoboHarness and TGL are test-time systems too (Table XI).

Base $\pi_{0.5}$ behind the same rewriter scores 88.23 %, a change of $+2.52$ $[1.14, 4.01]$ points. The rewriter alone, with no training of ours, gives the base policy that much. On base $\pi_{0.5}$, we detect no difference between the rewriter and the

closed-set mapping ($+0.15$ $[-0.14, 0.45]$ points).

XII. Other policies

Figure 8 shows the three policies without and with our data, and Figure 9 shows them on each factor.

We train OpenVLA [16] and FLUX-3 Action [18] on our full mix. It has the shares of the camera stage of the closed-set model (mxri-lg-va-1), and the same rule removes views near a test camera. OpenVLA reads the prompts of the vocabulary-guided rewriter. Our first version gave it a closed-set stage instead (Appendix C). FLUX-3 has a frozen text encoder, so it gets the reworded instructions as a part of its training mix. Both use the cleaned file of reworded instructions.

Each comparison is a pair, with one seed, one protocol and one host type. These policies use their own evaluator settings, so their scores do not compare with the $\pi_{0.5}$ rows. Appendix C gives the shares and the training details of each pair.

Across the three policies, the full mix improves robot start. Only $\pi_{0.5}$ has the ablation that removes the recovery clips alone, so we do not credit the clips with the gain of the other policies.

FLUX-3. We compare our full mix against FLUX-3 fine-tuned on the LIBERO-Plus training set only. Both runs start from the FLUX-3 Action base checkpoint (Hugging Face revision 62878e29). They use one trainer with the same settings, and the two recipes differ only in their data. We score both at their last step (4,000). At test time, both runs read the benchmark instruction as built, with its file-name text, and no rewriter changes it.

The first run of the baseline diverged near update 1,750, and our baseline resumes it from step 1,500 (Appendix C). The run with our mix trained once, without divergence.

Against the baseline, our mix changes the seven-axis mean by $+0.57$ $[-3.00, 4.08]$ points, with intervals from the 40 tasks. That interval contains zero. Per factor, in points: camera -2.44 $[-6.67, 1.84]$, robot start $+12.71$ $[8.10, 17.10]$, language -0.39 $[-4.45, 3.78]$, light $+1.31$ $[-3.81, 7.20]$, background -6.23 $[-12.17, -0.64]$, sensor noise -0.81 $[-4.74, 2.72]$, layout -0.13 $[-4.97, 4.25]$.

OpenVLA. The baseline is the published OpenVLA-OFT+ checkpoint of the benchmark authors, which they fine-tuned on the LIBERO-Plus training set (Hugging Face revision a85655ec). We score it, and then we continue to train it on our full mix. Thus the start, the steps and the data all differ from those of the authors. This pair measures our continuation of their fine-tune, so we read it as exploratory. Both models read the instructions that the vocabulary-guided rewriter of Section VIII writes, so the two read the same prompt on every case.

Both models run the protocol of the OpenVLA-OFT evaluation code. It executes all 8 actions of each query and snaps the gripper command to ± 1. We have no repeat pass of an OpenVLA model, so we read this pair by its intervals only. They cover the sampling of the cases, not the noise between two passes, which is 0.44 points for $\pi_{0.5}$ and not known for OpenVLA.

TABLE IX
OpenVLA-OFT+ before and after our training data

Factor	OFT+	+ our data	Change
Camera	93.7	92.9	-0.8 [−4.6, 3.2]
Robot start	28.4	88.4	$+60.0$ [54.9, 64.8]
Language	90.1	87.5	-2.6 [−10.9, 5.7]
Light	97.6	87.8	-9.8 [−19.2, −1.9]
Background	96.6	92.2	-4.4 [−11.9, 1.8]
Sensor noise	95.0	92.1	-2.9 [−8.8, 2.7]
Layout	74.0	77.3	$+3.3$ [1.1, 5.8]
Seven-axis mean	82.2	88.3	$+6.1$ [2.1, 9.8]
Total	81.0	88.2	$+7.2$ [3.3, 10.8]

Success, % of the cases of each factor, under our protocol, with the vocabulary-guided rewriter on both sides. OFT+: the checkpoint that the benchmark authors published. + our data: OFT+ further trained on our full mix. Change: points, with 95 % intervals from the 40 tasks.

Against OFT+, our continuation changes the seven-axis mean by $+6.12$ $[2.09, 9.82]$ points and the total by $+7.20$ $[3.32, 10.80]$ points (Table IX). The gain comes from robot start, $+60.00$ $[54.85, 64.78]$ points, and layout, $+3.34$ $[1.11, 5.85]$ points. Light changes by -9.81 $[-19.19, -1.90]$ points. Language changes by -2.60 $[-10.93, 5.66]$ points, so this pair shows no language gain from our training.

The first pair. Our first version scored the same two checkpoints without the rewriter: our continuation with its closed-set stage, and OFT+ on the raw prompt. That pair changes the seven-axis mean by $+7.52$ $[3.98, 10.73]$ points, but it mixes our training with prompt handling. Under our protocol, OFT+ scores 79.82 % on its raw prompts, against the 79.6 % that its authors published (Appendix A).

On our continuation, we detect no difference between the vocabulary-guided rewriter and the closed-set stage: the change is -0.29 $[-0.81, 0.22]$ points. On OFT+, the rewriter changes the seven-axis mean by $+1.12$ $[-1.13, 3.04]$ points, an interval that contains zero. We registered none of the rewriter pairs, so we read them as measurements.

Like the public training code of OpenVLA-OFT [17] (`vla-scripts/finetune.py`, lines 266, 388 and 625–640 [15]), our continuation keeps the action head of the first GPU. It trains with a batch of 64, and the authors used 16 (Appendix C).

XIII. Limits and the next skill

Object layout. Layout is a skill that we found but did not fix. The leaderboard system changes it by -1.44 points against base $\pi_{0.5}$ (Figure 1 gives the interval), and we did not split its layout cases by kind. The analysis below reads the closed-set recipe (mxri-lg-2). Its layout cases are of two kinds: scenes with added objects (829 cases) and scenes where every object moves (696 cases). The recipe succeeds on 95.66 % of the first kind but on only 73.28 % of the second, which holds 186 of its 222 layout failures. On the moved scenes, success falls by 18.0 points from displacement level 1 to level 5, levels that come from the case names. There, libero_goal is the weakest suite: the recipe succeeds on 34.59 % of its cases.

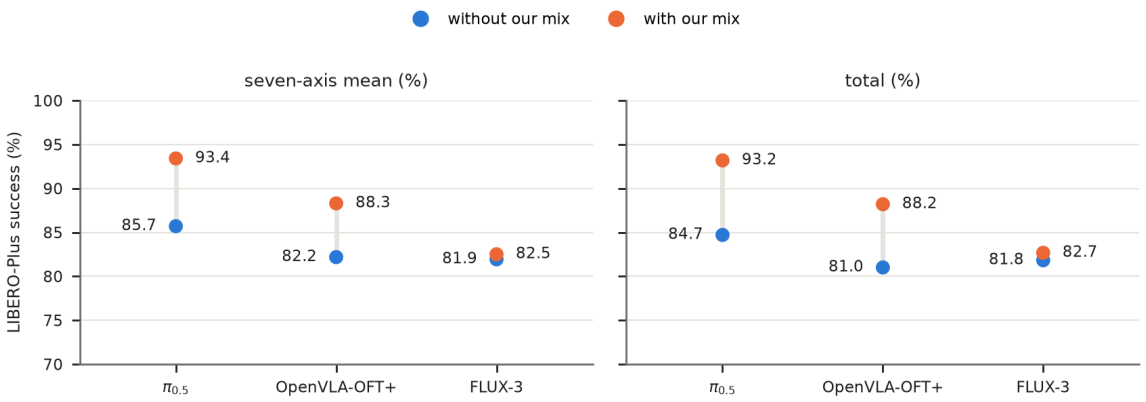


Fig. 8. Three policies without and with our data mix, under our protocol. Left: the seven-axis mean. Right: the total. $\pi_{0.5}$: base $\pi_{0.5}$ and the leaderboard system (the mean of two passes); its change also holds the vocabulary-guided rewriter (Table V). OpenVLA-OFT+: the benchmark authors' checkpoint, and that checkpoint further trained on our full mix, both with the vocabulary-guided rewriter. FLUX-3: FLUX-3 fine-tuned on the LIBERO-Plus training set only, and on our full mix. Each pair differs in its own way (Sections X and XII).

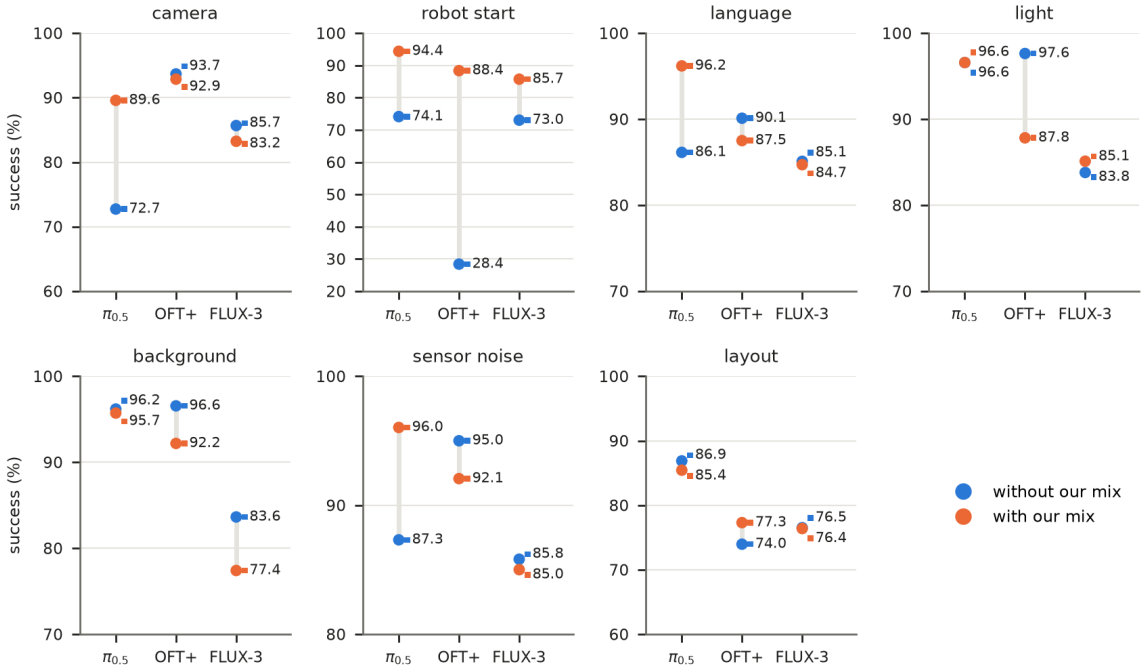


Fig. 9. The policies of Figure 8 on each perturbation factor, without and with our data mix, under our protocol. Success, % of the cases of that factor. Each panel has its own vertical scale. OFT+: OpenVLA-OFT+.

Our three layout arms changed the moved scenes by at most 0.43 points against the closed-set recipe, with standard errors of about 1.2 points. Their data barely covered the failing cells: 16 % of the episodes of one arm fell in them. One LIBERO-10 task, the black book in the caddy, can succeed only while the book moves. At rest, the book is 3.48 mm below the goal region.

What our data took from the benchmark. We built the data sets for the perturbation types that the benchmark paper publishes (its Appendix A) [9]. The robot-start range is the published one, and the camera ranges span the test poses' own values, up to 1.3 times the largest. The rewording styles follow the published types, and their author read the test instructions before writing them (Appendix F).

The closed-set language stage of each $\pi_{0.5}$ model in the tables, except mxri-lg-3 (Table XVI), trained on all 13,546 pairs. The author of their style guides wrote them after seeing the test instructions, and 23 pairs are within 0.10 (normalised edit distance) of a test instruction (Appendix F). The example phrases of the guides account for at most 0.02 points of the seven-axis mean, or 0.09 with the broader stem. We did not train the stage again without them, because the gain does not

come from the close prompts (Table XV).

Later runs use a cleaned file of 10,684 pairs: the language stage of OpenVLA, the full mix of FLUX-3, and both new runs of the rewordings comparison (Appendix F). Our served policy changes each prompt before it acts, and other policies can handle prompts differently.

Repeats and overlap. We trained most models once and scored each of them in one pass. The camera stage has three training seeds, the closed-set recipe two training runs, and the start run and the leaderboard system two passes each (Section IV). We also compared the start states of the recovery clips with the test start states. Of the recovery episodes, 2.4 % start from the object placement of a test case, and we did not measure the effect (Section V).

We did not test the original LIBERO suites again, and we tested in simulation only.

The vocabulary-guided rewriter. The rewriter of the leaderboard system is a closed model, GPT-6-astra, behind an API. The open-weight Qwen3.5-9B costs 0.42 to 0.60 points of the seven-axis mean against it. We scored the leaderboard system in two passes, with no registered bar. Its rulebook holds the word list of the training instructions, so a new task needs its words in that list. Over two passes, it differs from the closed-set model by -1.09 $[-2.39, 0.12]$ points on camera, with a separate run of the camera stage.

Contamination of the rewriters. The LIBERO-Plus test instructions have been public on GitHub since 2025, so GPT-6-astra and Qwen3.5-9B may have seen them in pretraining. The author of our rulebook did not read them, but that does not rule this out: it says nothing about what the models learned before we used them. We probed Qwen for memorization. Given the first half of each of the 1,509 language test instructions, it completes 0.07 % of them word for word, against 0.00 % of 780 paraphrases that were never public. The difference, $+0.07$ $[0.00, 0.21]$ points with its interval from the 40 tasks, meets our registered bar for no suspected memorization; the probe detects verbatim recall, not weaker exposure. The same probe on GPT-6-astra has not run yet.

Other evaluators. We did not score the leaderboard system with vla-eval, the evaluator of the AllenAI leaderboard. On the closed-set model, vla-eval with our settings gave 93.62 %, and our protocol 93.23 % (Appendix B). That is one reading on another model, and we derive no correction from it.

The next step. Our rewriter is a side model, and our closed-set stage knows only 40 instructions. The next step is a policy whose own language model rewrites any instruction into the words of its training data. It would need no closed set and no side model. We leave this open, for us or for others.

Use of language models

Language models wrote the instruction pairs of Appendix F. Claude Sonnet 5 wrote the 11,539 rewordings, and a Claude Opus 5.5 workflow wrote 1,488 of the captions. Claude Opus 5.5 also wrote the 2,400 paraphrases of the arms without the mapping. GPT-6-astra and Qwen3.5-9B rewrote the test instructions for the vocabulary-guided rewriter of Section VIII, and Claude Sonnet 5.5 judged their held-out rewrites. We used a language model (Claude) to help write, check and format this paper and its analysis scripts. The authors checked each number against its source (Appendix E).

Data and licences

We use LIBERO [23], and LIBERO-Plus [9] and its training set. The licences that we found are these:

- LIBERO: code under MIT (its LICENSE file), and demonstrations under Apache-2.0 (their card).
- LIBERO-Plus: training set and assets under MIT (the metadata of their cards, with no LICENSE file). The cards do not state the licence of the meshes that the assets take from other sources. We found no licence for its code and test cases, and none for the LIBERO-Plus simulator fork in which we render our camera views.
- OpenVLA and OpenVLA-OFT: code under MIT. The OpenVLA weights derive from Llama 2 and fall under the Llama 2 Community License (the OpenVLA README). The OFT+ checkpoint's card says MIT, and its base is Llama-2-7b, so the Llama 2 terms apply too.
- $\pi_{0.5}$ and PaliGemma: the pi05_libero weights derive from PaliGemma and fall under the Gemma Terms of Use. The openpi code is under Apache-2.0.
- Qwen3.5-9B: weights under Apache-2.0 (its Hugging Face card).
- GPT-6-astra: we made its rewrites through a ChatGPT subscription, under OpenAI's terms for that service. We publish the rewrites under the notice of our release.
- FLUX-3 Action: weights under the FLUX Kommunity License v1.0, a non-commercial licence. Code and text encoder under Apache-2.0. We fine-tuned and evaluated it for non-commercial research only. We release none of its weights or fine-tunes, only the per-case success flags of our two passes.

Release

We release, under Apache-2.0, the per-case results, the adapter configuration, and our camera sampler and transform. We also release the rulebook and the rewrite tables of Section VIII, and the language files of Appendix F. The repository at https://github.com/Alakazam-studios/targeted-data-libero-plus also holds our render driver. It does not hold the renderer that the driver calls, so the driver does not run on its own. We release generators rather than rendered frames, because some assets in the LIBERO scenes restrict redistribution. The repository's README states what we release and why.

We release the weights of the leaderboard policy (mxri-va) as the main weights, and those of the closed-set model (mxri-lg-va-1) beside them. Both are under the Gemma Terms of Use, at https://huggingface.co/alakazamworld/pi05-libero-plus-targeted.

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TABLE X
OPENVLA-OFT+: ITS PUBLISHED SCORE AND OURS

Factor	Published	Ours	Difference
Camera	92.8 [91.4, 94.0]	93.7 [92.4, 94.8]	+0.9 [−0.9, 2.6]
Robot start	30.3 [28.1, 32.6]	27.6 [25.4, 29.9]	−2.7 [−5.9, 0.5]
Language	85.8 [84.0, 87.5]	84.6 [82.7, 86.3]	−1.2 [−3.7, 1.3]
Light	94.9 [93.5, 96.0]	96.5 [95.3, 97.4]	+1.6 [−0.1, 3.3]
Background	93.9 [92.3, 95.2]	97.2 [96.0, 98.0]	+3.3 [1.6, 5.0]
Sensor noise	89.3 [87.7, 90.7]	93.8 [92.5, 94.9]	+4.5 [2.6, 6.4]
Layout	77.6 [75.4, 79.6]	74.1 [71.8, 76.2]	−3.5 [−6.5, −0.5]
Total	79.6 [78.8, 80.4]	79.8 [79.0, 80.6]	+0.2 [−0.9, 1.3]

Each rate: % of the cases of that factor. Difference: ours minus published, points. Intervals: Wilson 95 % on each rate, and unpaired on the difference, with our case count of each factor. Published: Fei et al. [9], Table 2 (arXiv v3), row "Ours", and the benchmark README at 4976dc30, which agree on each factor. For the total, Table 2 gives 79.5 and the README and the text of the paper give 79.6. The seven values weighted by the number of cases give 79.56.

APPENDIX A
CALIBRATION OF OUR PROTOCOL

OpenVLA-OFT+ is the one model in this paper with a score that the benchmark authors published and a score from our protocol. We score the checkpoint at its Hugging Face revision a85655ec. When a difference is inside its interval, we did not detect a difference under this comparison. That is not agreement: showing agreement needs an equivalence margin. Table X gives the two scores per factor.

The case-weighted totals are 79.6 % published and 79.82 % under our protocol, and the difference is inside its interval. After a Holm correction over the seven factors, these differ beyond their intervals: sensor noise and background. Only at an uncorrected 95 % level: layout.

APPENDIX B
SCORES THAT OTHER PAPERS REPORT

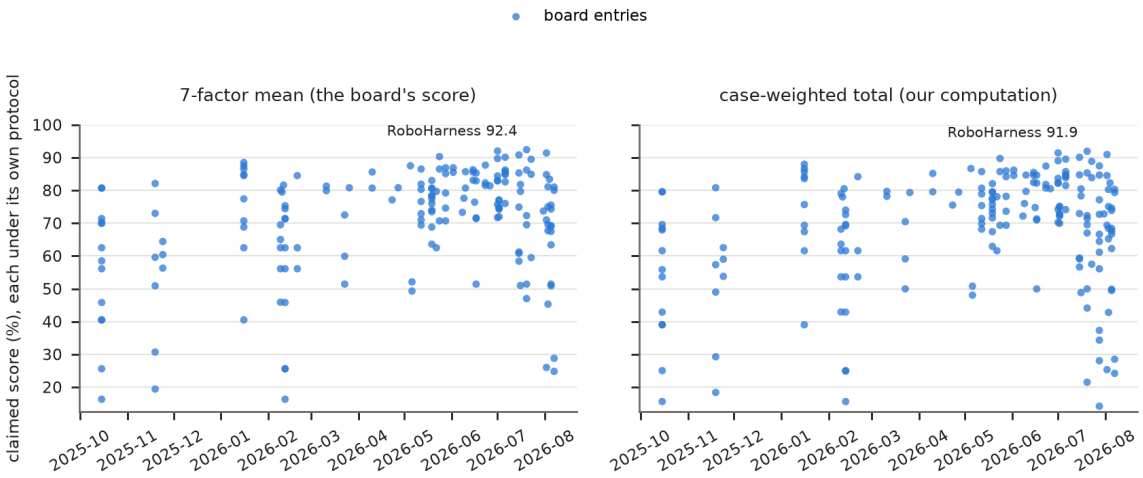


Fig. 10. Every LIBERO-Plus entry of the AllenAI leaderboard (its data at commit 41bd8a44, last updated 2026-08-10). We draw each claimed score against the date of the first arXiv version of its paper. Every point is the paper's own claimed number under its own evaluation protocol, and the figure does not draw our scores. Most papers do not state their settings, so the order is by claimed number, not a controlled comparison. Left: the board's score, the mean of the seven factors. Right: the case-weighted total, which we compute from the seven factor scores of each entry. The board mixes models trained on LIBERO only and models fine-tuned on LIBERO-Plus data, and we do not separate them.

Other papers. Figure 10 and Table XI give the LIBERO-Plus scores that other papers report. None of them states all its evaluation settings, so the order of Table XI is an order of reported scores, not a controlled comparison. In that order, the leaderboard system is above every score of other work, and so is the closed-set model (mxri-lg-va-1).

For context, we also scored the closed-set model with vla-eval [7], at commit 6cc3e1bd of its repository [2]. Its LIBERO-Plus support is in that repository, not in the paper. With our settings (5 flow steps, 5 actions per query, own decoding), it scores 93.62 %, and 93.35 % $[92.85, 93.82]$ (total). We derive no correction between evaluators from these numbers.

TABLE XI
LIBERO-PLUS SCORES THAT OTHER PAPERS REPORT

Method		Total	7-axis
This paper, leaderboard system	FT, TT	93.2	93.4
This paper, mxri-lg-va-1	FT	93.1	93.2
RoboHarness [13]	TT	91.9	92.4†
TGL [26]	TT	91.9†	92.4
PatchWAM [32]	FT	91.5†‡	91.9†
ZimaBlue [35]	FT	91.5	92.0
V-JEPA Policy [41]	ZS	91.5	91.9†
Qwen-RobotManip-Context‖	ZS	91.4	92.0†
Hermite-VLAReg [24]	FT-L	90.9	91.4†
Kairos-joint [14]	FT	90.9†	91.3†
CAC-VLA [37]	FT	90.4§	91.0†
QuoVLA [33]	FT	90.3†	90.9
QuoVLA [33]	ZS	89.7†	90.3
$\pi_{0.5}$ (base) [14, 24, 33]		85.7	86.5†
OpenVLA-OFT+ [9]	FT	79.6	80.7†

LIBERO-Plus scores, % of cases. Rule: every entry with a total or a seven-axis mean of at least 90 % that we found in published papers, as of 2026-10-12. Rows: sorted by total, with each paper's own numbers. PatchWAM's LIBERO-Plus table, which we also searched, differs from some of them. Below the rule: base $\pi_{0.5}$ and the benchmark authors' model. Base $\pi_{0.5}$: Hermite-VLAReg, QuoVLA and Kairos print the same row and do not say which checkpoint or evaluation settings they use. RoboHarness gives the same total from the pi05_libero checkpoint, with other factor scores. Kairos-joint: the paper prints 90.8 in its Average column, which is case-weighted in its other rows. Its factor scores give 90.86 weighted. Total: weighted by cases. 7-axis: the mean of the seven factors. †our computation from the factor scores that the paper prints. ‡PatchWAM prints 91.8, which matches neither aggregation of its factor scores. §CAC-VLA's own paper. PatchWAM's table gives 90.1 for it, in a Total column that mostly holds seven-axis means. OpenVLA-OFT+: the benchmark README's name for the row "Ours" (+ PT) of its paper's Table 2. FT: fine-tuned. FT-L: fine-tuned, but the paper names no LIBERO-Plus training data (our reading). ZS: trained on LIBERO, with no LIBERO-Plus training. TT: test-time system. RoboHarness: its paper reports the total, an average weighted by the official instance counts, and gives no trial or seed counts. Its 7-axis value is our mean of its seven columns. Its task-and-motion planner works from simulator object poses with 5 to 10 % added noise. The leaderboard system: the mean of two passes. mxri-lg-va-1: one pass. ‖as ABot-M0.5 [6] reports it. Anchor-Align VLA [8] reports LIBERO-Plus on the libero_spatial suite only (its Appendix C.3), so we removed its row. It prints 90.3 for that suite. RV-ICL [3] reports 95.8 % on a subset of 120 cases, not on the benchmark, so it is not in the table. Different evaluators: an order of reported scores, not a controlled comparison.

APPENDIX C
OTHER POLICIES IN DETAIL

Data. OpenVLA draws its training windows from the LIBERO-Plus training set (44.55 %), the recovery clips (22.45 %) and the rendered camera views (33 %). These are the shares of the camera stage of mxri-lg-va-1, with the same rule that removes views near a test camera. FLUX-3 has the same shares, but 15 points of the training-set share go to a copy with reworded captions: 29.55 % plain, 15 % reworded. From the counts of its build, its draws are 29.63 % (plain), 15.00 % (reworded), 22.38 % (recovery clips) and 32.98 % (camera views). CLAIMS.md (item 65) points to each lane's specification.

The language data enters each policy in a different way. In our first version, OpenVLA got our closed-set stage on its own language model. Its pair in Section XII reads the rewriter's prompts.

FLUX-3 has a frozen text encoder and no language-model head, so it gets the reworded instructions as a separate part of its training mix. This part is a copy of the LIBERO-Plus training set with reworded captions. It takes 15 % of the training windows from the share of the training set. We set the 15 % from the share of language cases in the test set (1,537 of 10,030, 15.3 %). We did not tune it.

Both policies use the cleaned file of reworded instructions. FLUX-3 also predicts video from one current frame. Our rendered views hold one camera pose per frame, so on windows from the views it trains the actions only.

The FLUX-3 baseline. The first run of the baseline diverged near update 1,750. Its recipe has no gradient clipping. Our baseline resumes that run from its step 1,500 with the same position, learning rate and optimizer state. On the same batches in the same order, the resumed run did not diverge.

Training is not bit-deterministic: 1 of 200 replayed updates matched bit for bit. So one replay cannot separate a rare numerical event from an effect of the data, and we do not know the cause.

The OpenVLA continuation. The public training code of OpenVLA-OFT calls its action head outside the data-parallel wrapper. Thus each GPU trains its own head on its share of the batch. The checkpoint keeps the head of the first GPU, which is an open issue of that code.

We infer, but did not measure, that OFT+ has the same property. Its code is that public code, and nobody published its exact training script. We kept that code unchanged, which does not make our run match theirs. Our continuation trains on 8 GPUs with the same code, so our checkpoint also holds the head of the first GPU. That head trained on one eighth of each batch.

The authors trained OFT+ with an effective batch of 16 [9], and our continuation uses a batch of 64.

APPENDIX D
INDEX OF CONFIGURATION

Evaluation. We use LIBERO-Plus at commit 4976dc30 and MuJoCo 3.2.3, with one trial per case and environment seed 42. The hosts have one L4 GPU, and the two passes of the start run also used some hosts with four GPUs. Our evaluator is protocol ee3f0803. The policy uses 5 flow steps and 5 actions per query, and robosuite 1.4 uses only the sign of its raw gripper command. A model with a language stage maps each instruction with its own language model.

We fixed these two settings when we designed the protocol on 2026-10-01, before we scored this paper's models with it. Five actions per query is the LIBERO default of openpi. Five flow steps is the value in the $\pi_{0.5}$ model configuration of RLinf (`pi0_5.yaml`) [29]. The default of openpi is 10.

We tuned neither setting on the test set. On a full libero_spatial pass of base $\pi_{0.5}$, 10 flow steps gave the same success (87.2 %). A sweep on held-out cases, registered before its score, found 10 flow steps no better (-0.60 points, with a paired standard error of 0.70 points). The protocol applies both settings to every model in the same way.

On AllenAI's code, mxri-lg-va-1 scored 1.91 points higher with 5 and 5 than with 10 and 10. Both changed together, so we do not know which one gives the gain.

Recovery clips. After the 10 settle steps, the script moves the gripper back to the pose of the recorded start state. It acts in the end-effector action space of the policy, with the gripper open. It stops within 0.01 m and 0.05 rad, or after 60 steps.

A DAgger-trained $\pi_{0.5}$ model (weights sha256 prefix be8ce1d9) then does the task. That model also trained on its own rollouts from the same LIBERO start states, and on one caption that equals a test instruction. Neither reached the data of mxri-lg-va-1: the clips carry plain task instructions, and our screen kept that caption out of the language file. We keep an episode only if the goal check of LIBERO succeeds. That gives 6,480 episodes, from three seeds of 2,000 starts each (95 % success) and a fourth seed that stopped early. Our training runs hold out the clips of one start per task (160 episodes) for the validation loss, but the start run held out none.

Training settings. Table XII gives the settings that the text does not.

TABLE XII
TRAINING SETTINGS

Stage	What trains	Settings
Camera stage (mxri-lg-va-1)	LoRA adapter, rank 16, on the MLP layers of the 27 SigLIP blocks (4.7 million parameters)	2,000 steps; one third of each batch from the views; cost 13.2 USD
Language stage ($\pi_{0.5}$)	the language model only	636 steps
OpenVLA continuation	from OFT+	25,000 steps, batch 64, 8 GPUs
FLUX-3 (both runs)	the whole model	4,000 updates, learning rate 1.92×10^{-4} ($\times 5$ for the action heads), weight decay 0.05, no gradient clipping

Models. Table XIII gives the models with plain names, the no-views arm and the camera arms. The validation loss is the held-out flow-matching error at the last step of the last action stage. We give it per data group: held-out demonstrations of the LIBERO-Plus training set, the recovery clips and the rendered views. All runs use one split file (p4c-split.json, 9d70d969). Their drawn validation frames can differ, so the losses of two rows are not on identical frames.

Each loss is the mean over a fixed sample of 2,040 to 2,048 held-out frames, through the training preprocessing (with augmentation). The rows use 7 different samples, so compare two rows only with care: mxri-lg-va-1 and mxri-lg-ad, for example, are on different frames. The models built on the start run saw these frames in its training. A language stage after the last action stage has no such loss.

Two runs show "not logged": the start run ran no validation pass, and the run with all rendered views predates our validation logging. For the start run, its learning rate, batch and W&B run are not in our records. The two arms with teacher-relabelled data record only their target shares (22.5 and 16.2 %). They have no validation group for that data. stats/config-index.json gives the source of every field.

TABLE XIII
TRAINING FACTS OF THE MAIN MODELS AND THE CAMERA ARMS

Model	Weights	Start	Stages (what trains, steps)	Validation loss, last action stage
base $\pi_{0.5}$	8a240bac	–	public checkpoint, not trained by us	–
mxri-lg-1	854ea3d0	base $\pi_{0.5}$	all weights 10,000	not logged
mxri-2	6bc90f19	base $\pi_{0.5}$	all weights 10,000	0.0105 / 0.0155 (plus/clips)
mxri-lg-2	5a416850	base $\pi_{0.5}$	all weights 10,000 → language stage 636	0.0105 / 0.0155 (plus/clips)
mxri-va (the leaderboard policy)	6ac6926e	mxri-2	adapter 2,000	0.0093 / 0.0155 / 0.0168 (plus/clips/views)
mxri-lg-va-1 (the closed-set model)	f32a5737	mxri-lg-1	adapter 2,000	0.0091 / 0.0152 / 0.0164 (plus/clips/views)
mxri-lg-ad	7264b6d5	mxri-lg-1	adapter 2,000	0.0094 / 0.0140 (plus/clips)
mxri-lg-vaf	12483fd4	mxri-lg-1	adapter 2,000	not logged
adapter first, then full fine-tune	9fad47c1	base $\pi_{0.5}$ (converted, f73267f7)	adapter 2,000 → all but vision 10,000 → language stage 636	0.0101 / 0.0135 (plus/clips)
full fine-tune first, then adapter	447d0e4b	mxri-2	adapter 2,000 → language stage 636	0.0091 / 0.0153 / 0.0163 (plus/clips/views)
rendered views, all weights	a4a5a9ba	base $\pi_{0.5}$	all weights 10,000 → language stage 636	0.0103 / 0.0147 / 0.0168 (plus/clips/views)
teacher-relabelled rollouts	25487eae	base $\pi_{0.5}$	all weights 10,000 → language stage 636	0.0108 / 0.0158 (plus/clips)
teacher-relabelled failures	cd378a61	base $\pi_{0.5}$	all weights 10,000 → language stage 636	0.0109 / 0.0157 (plus/clips)
views and layouts	8e0332da	base $\pi_{0.5}$	all weights 10,000 → language stage 636	0.0106 / 0.0139 (plus/clips)
layouts from moved cameras	044c95e7	base $\pi_{0.5}$	all weights 10,000 → language stage 636	0.0104 / 0.0137 (plus/clips)

Weights (sha256 prefix), start, training stages and validation loss. The rows are the models with plain names (Section IV), the no-views arm and the camera arms of Table IV. Table II names every $\pi_{0.5}$ model that the text uses, and Section IV counts every model scored on the test set.

TABLE XIV
SOURCES OF THE INSTRUCTION PAIRS

Part	Pairs	Writer	Test instructions in its input
Rewordings	11,539	Claude Sonnet 5, no tool access	no (its prompt holds the 40 task instructions and a style list that its author wrote after seeing them; one example phrase of the list occurs in 7 test instructions)[a]
Captions	1,607	1,488 from a Claude Opus 5.5 workflow, 119 by hand	no; its author wrote its style guide after seeing them, and two of its example phrases occur in test instructions
Identity	400	the 40 task instructions, 10 times each	not applicable

[a] Its five styles of rewording follow the three rewording types that the LIBERO-Plus paper publishes (its Appendix A.6) [9].

APPENDIX E
CLAIM TO ARTEFACT

The file CLAIMS.md of our repository (Release) lists every number of this paper with its public evidence, or marks it an internal record. The per-case results in that repository recompute every score of our protocol. The item numbers that the text cites are the rows of that file.

APPENDIX F
A CLOSED-SET LANGUAGE STAGE

The data. A language model (Claude Sonnet 5) wrote 11,539 rewordings of the 40 instructions. With 1,607 captions and 400 identity pairs, this gives 13,546 instruction pairs.

We checked the pairs against the test prompts (Table XIV). No pair is equal to any of the 10,002 distinct test prompts as the benchmark serves them. Before the check, we lowercase both and turn every character other than a letter or a digit into a space. We measure how close a pair is to a test instruction by the normalised edit distance. It is the number of single-character edits (Levenshtein) that turn one instruction into the other, divided by the length of the longer one.

The 400 identity pairs are LIBERO's 40 task instructions, the ones that its training demonstrations carry. No test prompt is equal to one. The benchmark serves the 8,493 cases of the six other factors with file-name text after the instruction, and the 1,537 language cases with reworded instructions.

The agents that wrote the training rewordings received only the 40 task instructions and a style guide, and read no files. Their author wrote the style guides after reading the test instructions. The guides follow the three rewording types that the benchmark paper publishes. Their author copied one example phrase from a test instruction, and two more also occur there. The three appear in 11 of the 1,537 language cases, and 2 of those gained from the language stage of mxri-lg-2. So they account for at most 0.02 points of the seven-axis mean, or 0.09 points with the broader stem "the appliance used for".

Apart from the 40 task instructions, the training file (sha256 bdb4dd12) holds none of the 1,509 distinct language test instructions.

Within an edit distance of 0.30, bound included, the captions hold 11.8 % of their pairs and the other rewordings 8.4 %. That fits style guides whose author read the test instructions before writing them.

Prompts far from any training pair carry the language gain (Table XV). The gain grows with the distance from the training pairs, also at the same level mix. The cases more than 0.30 from any training pair gain 15.4 points, and the 24 closest cases were already at 100 %.

TABLE XV
LANGUAGE CASES BY DISTANCE TO THE NEAREST TRAINING PAIR

Distance	Cases	mxri-2	mxri-lg-2	Change	Level-adjusted
≤ 0.10	24	100.0 [86.2, 100.0]	95.8 [79.8, 99.3]	-4.2 [-12.2, 3.8]	-5.3 [-14.4, 3.7]
0.10–0.30	483	93.8 [91.3, 95.6]	98.3 [96.8, 99.2]	$+4.6$ [2.4, 6.7]	$+5.0$ [2.4, 7.5]
> 0.30	1,030	80.9 [78.4, 83.2]	96.3 [95.0, 97.3]	$+15.4$ [12.8, 18.0]	$+13.8$ [11.5, 16.1]

Language-factor cases by the distance from the test prompt to the nearest training pair (identity pairs excluded). Success, % of the cases of the band, Wilson 95 % intervals. Change: mxri-lg-2 minus mxri-2, one run before and after its language stage, points, paired by case. Level-adjusted: the change within each level, weighted by the level mix of all 1,537 cases. We named these bands before we read them (not committed before the reading), and the reading is descriptive.

TABLE XVI
THE LANGUAGE STAGE ON THE CLEANED FILE

Factor	mxri-3	mxri-lg-3	Change
Camera	80.2	84.8	$+4.6$ [1.6, 7.9]
Robot start	92.3	95.0	$+2.7$ [0.6, 4.9]
Language	84.5	95.1	$+10.6$ [5.1, 16.4]
Light	95.8	96.3	$+0.5$ [-0.8, 1.7]
Background	95.5	96.1	$+0.6$ [-0.5, 1.6]
Sensor noise	92.7	95.4	$+2.7$ [1.3, 4.1]
Layout	85.0	85.8	$+0.8$ [-0.8, 2.4]
Seven-axis mean	89.4	92.6	$+3.2$ [2.0, 4.4]
Total	88.9	92.3	$+3.4$ [2.2, 4.7]

mxri-3: $\pi_{0.5}$ fine-tuned on the demonstrations and the recovery clips. mxri-lg-3: mxri-3 with the language stage on the cleaned file. Success, % of the cases of each factor. Change: points, with intervals from the 40 tasks.

The stage. One stage trains only the language model of the policy. It rewrites each instruction into a task instruction, token by token, before the policy acts. Its training outputs are only the 40 task instructions, so it uses the fact that the benchmark has 40 tasks. The mapping is correct for 1,489 of the 1,537 language cases (96.88 %) in our pass of mxri-lg-va-1. The count is the same in our pass of the start run, which has the same language model. On the other 8,493 cases, it sends the case's own task instruction every time.

Of the 48 wrong language cases, 39 decode to another of the 40 instructions. The other 9 decode to none of them, for example "pick up the black bowl on the tree and place it on the plate".

A cleaned file gives a similar gain. Our later runs use a cleaned file of 10,684 instruction pairs. It drops every pair within an edit distance of 0.30, or a word-set Jaccard of 0.60, of a language test instruction, and all 1,607 captions. We dropped the captions when we believed that their writers had the test instructions. The run records later showed that they did not. On the same recipe, its language stage changes language by $+10.61$ $[5.14, 16.42]$ points and the seven-axis mean by $+3.21$ $[2.01, 4.44]$ points (Table XVI).

The cleaned file is not free of near-test wording. Of its 10,684 rows, 1,887 (17.7 %) are near a language test instruction. Near means an edit distance of at most 0.35, or a word-set Jaccard of at least 0.50. For 786 of the 1,537 language test cases (51 %), the file holds such a neighbour.

The end of this appendix shows that the language gain of this stage is its mapping onto the 40 tasks (Table XVIII).

What we tested. Without the stage, the closed-set recipe loses 11.71 points on language $[6.63, 17.10]$ and 3.02 points on the seven-axis mean (Table VIII). The intervals come from the 40 tasks. It also loses 4.69 points on camera and 2.32 points on robot start.

Without the stage, the policy gets the file-name text in its prompts on those six factors. Thus the camera and robot losses can come from the prompt defect. This row cannot separate the language skill from the handling of that defect.

Another idea did not help: a classifier for the instruction failed its first check (0.60 accuracy against a bar of 0.70).

Rewordings in the fine-tuning mix. We also tried the rewordings as part of the fine-tuning mix, in place of a language stage. Three runs share one recipe: 10,000 steps from base $\pi_{0.5}$ on the LIBERO-Plus training set and the recovery clips. In mxrirw, reworded copies of the LIBERO-Plus episodes take 15 % of the draws. The model mxri-lg-3 adds the language stage, and mxri-3 adds neither. Against mxri-3, the rewordings change language by $+1.95$ $[0.40, 3.69]$ points and the seven-axis mean by $+0.93$ $[0.31, 1.59]$ points, with intervals from the 40 tasks.

Against mxri-lg-3, they change language by -8.65 $[-14.02, -3.61]$ points and the seven-axis mean by -2.28 $[-3.38, -1.13]$ points. On this recipe, the language stage gives the gain of Table XVI. The model mxri-lg-3 maps each instruction with its

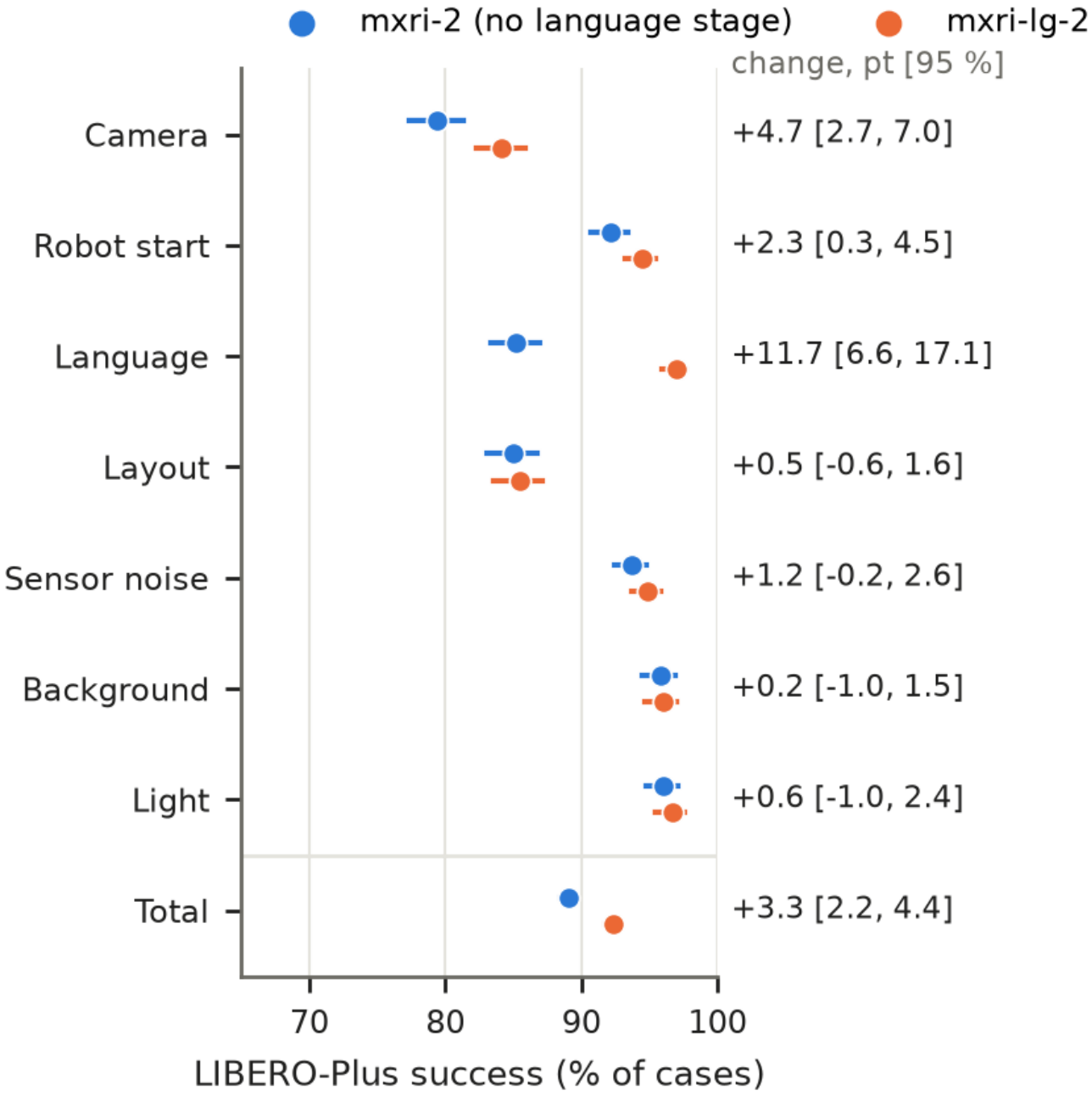


Fig. 11. Skill: Language. The models mxri-2 and mxri-lg-2, before and after the language stage. Without the stage, the policy also gets the file-name text that the benchmark code adds to the prompts. So this pair also measures the handling of that defect. Success, % of the cases of that type, Wilson 95 % intervals. Right: the change paired by case, points, with its 95 % interval from the 40 tasks.

own language model before it acts. The other two read the raw prompt, which on six factors carries the benchmark's file-name text (Section VIII). So the language stage, with its mapping, gives more than the rewordings in the mix. We did not separate the two.

Paraphrases without the mapping. We then asked whether the policy can earn a language gain with no test-time mapping. Three arms continue the closed-set model (mxri-lg-va-1) with the same data, steps and seed. On each draw, a paraphrase replaces the instruction with probability 0.8 (arm A), 0.5 (arm C) or 0 (arm B, the step control). Claude Opus 5.5 wrote 2,400 paraphrases from the task instructions and one frame per task, blind to the test set. The cleaned-file rule kept 1,478 of them.

We registered the criteria before any score and scored each arm once, with the mapping off. On language, A gains $+2.60$

TABLE XVII
PARAPHRASES WITHOUT THE MAPPING

Pair	Language	Camera	Robot start	7-axis
A − B	$+2.60$ $[0.12, 5.47]$	$+0.75$ $[-0.94, 2.35]$	$+0.39$ $[-0.71, 1.52]$	$+0.34$ $[-0.43, 1.19]$
C − B	$+2.47$ $[0.46, 4.62]$	$+0.44$ $[-1.06, 2.01]$	-0.26 $[-1.44, 0.96]$	$+0.27$ $[-0.54, 1.04]$
A − C	$+0.13$ $[-1.28, 1.61]$	$+0.31$ $[-1.53, 2.22]$	$+0.65$ $[-0.72, 1.99]$	$+0.07$ $[-0.65, 0.76]$
A − mxri-lg-va-1	-8.26 $[-12.18, -4.53]$	$+0.06$ $[-1.67, 1.74]$	$+0.06$ $[-1.71, 1.59]$	-1.22 $[-2.11, -0.38]$
N − mxri-3	$+2.73$ $[-0.26, 6.23]$	$+2.31$ $[0.00, 4.71]$	$+1.10$ $[-0.85, 2.95]$	$+1.09$ $[0.24, 1.94]$
N − A	$+0.07$ $[-2.49, 2.40]$	-8.26 $[-11.90, -4.81]$	-1.68 $[-3.31, -0.32]$	-1.49 $[-2.23, -0.75]$

Arms A, C and B continue mxri-lg-va-1 with a paraphrase share of 0.8, 0.5 and 0, scored with the mapping off. N trains base $\pi_{0.5}$ on the recipe of mxri-3 with a share of 0.8, and its registered primary is N − mxri-3 on language. mxri-lg-va-1: with its mapping. Change in points, paired by case, with intervals from the 40 tasks.

TABLE XVIII
WHERE THE GAIN OF THE CLOSED-SET MODEL COMES FROM

Change	Source	Robot	Lang.	Camera	7-axis
remove the file-name text	benchmark	$+3.35$ [0.64, 6.26]	-0.20 [−0.44, 0.00]	$+4.94$ [1.97, 7.98]	$+1.07$ [0.14, 2.08]
map each instruction to one of the 40 tasks	benchmark	$+0.06$ [−0.20, 0.33]	$+9.30$ [3.67, 15.01]	-0.13 [−0.48, 0.19]	$+1.29$ [0.45, 2.16]
our training, on identical prompts	training	$+17.42$ [14.04, 20.83]	$+0.20$ [−1.85, 2.66]	$+13.13$ [7.80, 18.90]	$+5.16$ [3.49, 6.92]
all three: mxri-lg-va-1 against base $\pi_{0.5}$	both	$+20.84$ [16.39, 25.59]	$+9.30$ [4.27, 14.63]	$+17.95$ [11.43, 24.97]	$+7.52$ [5.65, 9.52]
mxri-2: remove the file-name text	benchmark	$+3.35$ [1.48, 5.25]	$+0.07$ [−0.13, 0.28]	$+4.13$ [1.45, 7.04]	$+1.13$ [0.45, 1.80]
mxri-2: the vocabulary-guided rewrite, beyond that	rewriter	$+0.06$ [−0.44, 0.74]	$+10.08$ [5.41, 15.11]	$+0.00$ [−0.63, 0.71]	$+1.43$ [0.73, 2.18]

Each row changes one thing and keeps the rest. Rows 1 and 2: base $\pi_{0.5}$, its weights unchanged, with the file-name text removed from its prompts, then with the prompts that mxri-lg-va-1 acts on. Row 3: mxri-lg-va-1 against base $\pi_{0.5}$, both on those identical prompts. The three rows add up to the fourth one. Last two rows: mxri-2, its weights unchanged, with the file-name text removed, then with the vocabulary-guided rewrite on top of that. They do not add to the rows above. Change in points, paired by case, with intervals from the 40 tasks.

$[0.12, 5.47]$ points over B, and C gains $+2.47$ $[0.46, 4.62]$ points (Table XVII). Between C and A the change is $+0.13$ $[-1.28, 1.61]$ points, so the gain does not grow past a share of 0.5. On camera and robot start, we detect no difference between A and B. A reaches 87.18 % on language and 92.01 % on the seven-axis mean. Against the closed-set model with its mapping, A changes language by -8.26 $[-12.18, -4.53]$ points, so paraphrases earn back about a quarter of the mapping's gain.

A fourth arm, N, starts from base $\pi_{0.5}$ instead: 10,000 steps on the recipe of mxri-3, with a paraphrase share of 0.8. Its registered primary criterion is *not met*: against mxri-3 it changes language by $+2.73$ $[-0.26, 6.23]$ points, and that interval contains zero. Its seven-axis mean changes by $+1.09$ $[0.24, 1.94]$ points, to 90.51 %. Against A we detect no language difference, and it changes the seven-axis mean by -1.49 $[-2.23, -0.75]$ points, almost all of it on camera. We started N's pass before we computed A's interval, its registered trigger. That interval then met the trigger.

Where the gain of the closed-set model comes from. The closed-set model and base $\pi_{0.5}$ do not read the same prompt. The language stage rewrites each instruction, and for 9,982 of the 10,030 cases it gives the case's own task instruction. On the six other factors, this also removes the file-name text that the benchmark code adds (Section VIII). The next paragraph measures what each of these changes is worth.

We scored base $\pi_{0.5}$ twice more, with the prompts changed and its weights unchanged. With the file-name text removed, it changes the seven-axis mean by $+1.07$ $[0.14, 2.08]$ points. The prompts that the closed-set model acts on also replace most instructions by their task instruction. With them, it changes by a further $+1.29$ $[0.45, 2.16]$ points.

On those identical prompts, the closed-set model gains $+5.16$ $[3.49, 6.92]$ points over base $\pi_{0.5}$, with intervals from the 40 tasks: camera $+13.13$ $[7.80, 18.90]$ and robot start $+17.42$ $[14.04, 20.83]$. The first two changes come from the structure of the benchmark. The last one is what our training adds.

On language, the closed-set model gains only $+0.20$ $[-1.85, 2.66]$ points on those prompts. So the language gain of the closed-set stage is its mapping onto the 40 tasks. The vocabulary-guided rewriter has the word list of the 40 instructions but no mapping onto them. We detect no language difference between it and the closed-set stage (Section VIII).

Removing the file-name text gains base $\pi_{0.5}$ $+4.94$ $[1.97, 7.98]$ points on camera alone. The text reaches every evaluation that sends the benchmark's task instruction as built, as the LIBERO client of openpi does. We measured its cost on $\pi_{0.5}$ models only.

APPENDIX G
A CROSS-VIEW LOSS

We added a cross-view loss to the adapter stage of mxri-lg-va-1. It pulls together the features of two views of one scene. Against mxri-lg-va-1, it changes camera by -11.69 $[-17.03, -6.85]$ points and the seven-axis mean by -17.57 $[-22.94, -12.79]$ points (intervals from the 40 tasks). Its seven-axis mean, 75.66 %, is below that of the start run, from which it starts (92.43 %).

We also trained all weights except the language model with this loss, and no adapter. Against mxri-lg-va-1, this arm changes camera by -6.50 $[-10.04, -3.27]$ points and the seven-axis mean by -2.13 $[-3.07, -1.27]$ points, with intervals from the 40 tasks. It differs from mxri-lg-va-1 in five ways, so it is not a one-factor test. This tests cross-view consistency continued from the start run for 2,000 updates with one-third pair data, not the regime of [12].

The cause. The weights pipeline is not the cause: the save, the export and the evaluation inputs check out. The cause is the objective of the first arm. In its mode, images without augmentation reach the loss only through the cross-view term, never through the action loss. The adapter then made those images give the same output across views, at the cost of their actions, and the evaluation uses exactly such images.

The run's own curves show this. The cross-view loss fell to 0.6 % of its start value. The action loss on the plain view rose by 33 % from its mean over steps 500 to 599. Our validation loss ran on augmented images only, so it stayed level and did not show it. The second arm trains the action loss on both plain views, and its curves show no such collapse. We measured the signature, and the mechanism is our inference.

APPENDIX H
TEST USE, PASS BY PASS

Each pass of our protocol on test cases, with the decision that read its score. Choice: a recorded decision read the score. It picked a reported system (the v1 headline on 2026-10-05, the leaderboard headline on 2026-10-16) or a next experiment. Measurement: a record fixed its role before its score, and no decision read it. Repeat: a seed or a second pass, run to measure noise.

Not reported: in no table or sentence of this paper. Purpose not recorded: no record settles its role. Most choice passes also had their role fixed before their score. Our records quote the evidence for each row, and we share them on request. [a]stopped part-way and never read.

Pass (paper name)	Cases	Category	Decision that read it
p4c-k0 (mxri-lg-1)	10,030	choice: reported system	v1 headline
p4c-k1	10,030	repeat	–
arm-control (mxri-lg-2)	10,030	choice: reported system	v1 headline
arm-lc22	10,030	choice: reported system	v1 headline
arm-l22	10,030	choice: next experiment	layout follow-up
arm-e1-final	10,030	choice: reported system	v1 headline
arm-e2-final	10,030	choice: reported system	v1 headline
arm-b22	10,030	choice: reported system	v1 headline
arm-dc16	10,030	choice: reported system	v1 headline
arm-r22	10,030	choice: reported system	v1 headline
arm-rl22	10,030	choice: reported system	v1 headline
arm-fla (mxri-lg-vaf)	10,030	choice: reported system	v1 headline
arm-fla-disjoint (mxri-lg-va-1)	10,030	choice: reported system	v1 headline
arm-nolang (mxri-2)	10,030	choice: reported system	leaderboard headline
arm-pi05-libero (base $\pi_{0.5}$)	10,030	choice: next experiment	paraphrase experiment
arm-fla-disjoint-nobank (mxri-lg-ad)	10,030	measurement	–
arm-noclips-nolang (mx)	10,030	measurement	–
arm-noclips (mx-lg)	10,030	measurement	–
arm-rew1 (mxrirw)	10,030	choice: next experiment	paraphrase experiment
arm-ctlr1-s1 (mxri-3)	10,030	choice: next experiment	paraphrase experiment
arm-ctlr1-lang (mxri-lg-3)	10,030	measurement	–
arm-cvfla (mxri-lg-vacv)	10,030	choice: next experiment	cross-view investigation
arm-cvsib (mxri-lg-vwcv)	10,030	measurement	–
arm-fla-disjoint-s2 (mxri-lg-va-2)	10,030	repeat	–
arm-fla-disjoint-s3 (mxri-lg-va-3)	10,030	repeat	–
arm-openvla-base	10,030	other policy / calibration	–
arm-openvla-mix	10,030	other policy / calibration	–
arm-flux3-base	10,030	other policy / calibration	–
arm-flux3-mix	10,030	other policy / calibration	–
arm-ctlr1-s3	10,030	measurement	–

Pass (paper name)	Cases	Category	Decision that read it
arm-racr1-s3	10,030	not reported	–
arm-pi05-libero-strip	10,030	choice: next experiment	paraphrase experiment
arm-pi05-libero-mapped	10,030	choice: next experiment	paraphrase experiment
arm-fl-a	10,030	choice: next experiment	launch of arm N
arm-fl-b	10,030	choice: next experiment	launch of arm N
arm-fl-c	10,030	choice: next experiment	launch of arm N
arm-fln	10,030	measurement	–
llmrw-full-arm	10,030	choice: reported system	leaderboard headline
llmrw-full-strip	10,030	choice: reported system	leaderboard headline
llmrw-full-va-gpt	10,030	choice: reported system	leaderboard headline
llmrw-full-va-gpt-p1	10,030	repeat	–
llmrw-full-va-qwen	10,030	measurement	–
llmrw-openvla-base-gpt	10,030	other policy / calibration	–
llmrw-openvla-mix-gpt	10,030	other policy / calibration	–
llmrw-lang-nolang-gpt	1,537	choice: reported system	leaderboard headline
llmrw-lang-va-gpt	1,537	measurement	–
llmrw-lang-fla-gpt	1,537	purpose not recorded	–
arm-layout-p21-g1	1,525	not reported	–
arm-layout-p21-t2b	1,525	not reported	–
arm-layout-p21-t2f	1,525	not reported	–
arm-layout-p21-t2g	1,525	not reported	–
arm-layout-p21-t2h	1,525	not reported	–
llmrw stopped L2 (lang-nolang-qwen)	1,537[a]	not reported	–
llmrw stopped L3 (lang-nolang-hugo)	1,537[a]	not reported	–
llmrw stopped L4 (lang-pi05-gpt)	1,537[a]	not reported	–
llmrw stopped L5 (lang-pi05-qwen)	1,537[a]	not reported	–
llmrw stopped full-fla-gpt	10,030[a]	not reported	–